

**TITLE: THE ERP SOFTWARE**

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## **1. Abstract**

This report details the creation, deployment, and testing of a software tool called Enterprise Resource Planning (ERP). It begins with an explanation of why ERP systems are important, sets out what the project aims to achieve, reviews existing knowledge on the topic, and describes how the software was designed and put into action. It then states about how the software was designed, and developed, how languages and coding was implemented and elaborates about the database used to manage the ERP 3(stock management) data within one single platform followed by the section that includes testing of the system and what still needs work. The main focus is on making sure the software is easy to use, works well, and manage the data by avoiding complexities in data management.

## **2. Introduction: -**

In the rapidly evolving landscape of business operations, organizations are recognizing the imperative need for efficient, streamlined, and integrated solutions to manage their diverse processes. Enterprise Resource Planning (ERP) systems have emerged as indispensable tools, providing a comprehensive platform that consolidates various business functions into a unified ecosystem. As businesses strive for increased agility, productivity, and data-driven decision-making, the development of a robust ERP software becomes a strategic imperative. ([www.Microsoft.com](http://www.Microsoft.com), 2023)

### **2.1 About ERP (Enterprise Resource Planning)**

An ERP system is a set of business applications or modules, which links various business units of an organization such as financial, accounting, manufacturing, and human resources into a tightly integrated single system with a common platform for flow of information across the entire business. With the use of the Internet as a business medium, organizations can use the expanded version of ERP, to connect their internal business systems with the systems of customers and suppliers. The purpose of ERP is to enhance an organization's competitiveness by improving its ability to generate accurate and timely information for managerial decision making. A complete ERP suite also includes enterprise performance management, software that helps plan, budget, predict, and report on an organization's financial results. ERP systems tie together a multitude of business processes and enable the flow of data between them. By collecting an organization's shared transactional data from multiple sources, ERP systems eliminate data duplication and provide data integrity with a single source of truth. Today, ERP systems are critical for managing thousands of businesses of all sizes and in all industries. To these companies, ERP is as indispensable as the electricity that keeps the lights on. (*Beheshti, 2006*)

ERP systems are designed around a single, defined data structure (schema) that typically has a common database. This helps ensure that the information used across the enterprise is normalized and based on common definitions and user experiences. These core constructs are then interconnected with business processes driven by workflows across business departments connecting systems and the people who use them.

The concept of my software is driven from the above statement as the technology and software created is a better and simple version for the companies to record their data within a single platform.

## **2.2 Project Motivation**

My motivation begins from the desire to enhance operational efficiency, budget record, and provide a unified platform that fosters collaboration across departments. By implementing a robust ERP system, I aim to eliminate difference between variety of departments, reduce manual errors.

The inspiration behind selecting this topic stems from my firsthand experience working within a startup organization. During my tenure, the company was using a Linux-based operating system for record-keeping and operations. While the Linux system was robust and reliable, I identified a gap in the graphical user interface (GUI) software employed for these tasks.

The existing solutions were somewhat complex, requiring a certain level of technical expertise to navigate effectively. Recognizing the need for a more streamlined GUI interface software, I was motivated to embark on a project that addresses these issues. This project is driven by the aspiration to bridge the gap between powerful functionalities, ensuring a smoother and efficient management in business industry.

## **3. AIMS**

Develop a robust Enterprise Resource Planning (ERP) software with a graphical user interface (GUI) to meet the evolving needs of modern businesses. The software aims to consolidate diverse processes into a unified ecosystem, fostering efficiency, agility, and data-driven decision-making. Inspired by the demand for a streamlined GUI, the goal is to offer companies a simplified platform for comprehensive data recording, collaboration, and improved operational capabilities.

## **4. OBJECTIVES**

To make sure we achieve our big goals, initially the motive is to follow a plan step by step. Each part of the project, from thinking up ideas, making the GUI look nice, writing the code that runs things behind, to testing everything out, needs to be done carefully and precisely. Using right tools for each job to make sure everything turns out the way it is expected to be. This process will cover everything, starting from researching ideas, creating the user interface that people see and interact with, doing the coding that makes everything work, and finally, testing everything really well. Using the best tools at each step is super important to make sure the project is a success.

## **5. LITERATURE REVIEW**

The objective of these systems is to provide the right materials to the right location at the right time (*Gavidia, 2017*). ERP is an integrated information management system that streamlines business processes by processing information in real-time, centralizing data and eliminating operational redundancies. (*Kanet and Stößlein, 2010*)

## 5.1 Historical Background

The term ERP was first used in the 1990s by the Gartner Group, but enterprise resource planning systems actually have their roots deep in the manufacturing industry and can trace their history back to the 1960s.

At this time, manufacturers needed a better way to manage, track, and control their inventory. Basic software solutions, known as MRPs or Material Requirements Planning systems, were developed to meet their needs. These systems helped manufacturers monitor inventory and reconcile balances, as well as included very basic manufacturing, purchasing, and delivery functions.

In the 1990s, the first true ERP systems came into use. These systems further expanded beyond previous iterations' basic inventory control and manufacturing processes to include other departments and functions, such as accounting, finance, and sales. These systems set the stage for ERP solutions as we've come to know them today, by integrating multiple processes and departments into one system.

Over the next few decades, ERP systems became more advanced. In recent years, big data and AI have once again transformed ERP systems, making them more powerful and adaptive. Big data enables ERP systems to handle massive amounts of information, providing businesses with valuable insights into their operations. AI adds a layer of intelligence, allowing ERP systems to learn from data patterns and make informed predictions, and AI-driven automation streamlines repetitive tasks. ([www.geniuserp.com](http://www.geniuserp.com), 2020)

ERP systems continued to evolve throughout the '90s. One major breakthrough was the advent of cloud ERP, first offered by NetSuite in 1998. With cloud ERP, widely seen as an improvement over on-premises systems, businesses could access critical business data through the web from any device with an internet connection. Cloud solutions meant companies no longer needed to purchase and maintain hardware, reducing the need for IT staffers and leading to easier implementations. (McCue, 2020)

## 5.2 Methods Used

In the development of this software, which is rooted in web development and incorporates elements of SQL (**Structured Query Language**) (Microsoft Corporation, 14/11/2023). injection and .NET C#, integration of various methodologies from each sector has been employed. This strategic approach ensures a robust and secure web application, leveraging the strengths of web development practices, resilience against SQL injection vulnerabilities, and the versatile capabilities offered by the .NET C# framework. The combination of these methodologies not only enhances the software's functionality but also enhances its security measures, creating a comprehensive and flexible solution for modern business needs.

### 5.2.1 Integrated Web Application Development Methodology

The methodology above was deployed for the development phase of the software this includes certain measures that should take place during the process as mentioned below.

- **Requirement Gathering and Analysis**

To acknowledge the project's goals, and business requirements and to define functional and non-functional requirements for the web application. Further elaborated in **section 4.3**.

- **Planning**

A project plan that outlines the scope, timeline, resources, and milestones. Planning the application's architecture, including data models, and interactions. Further elaborated in **section 4.3, section 4.4**.

- **Design**

Design the user interface (UI) and user experience (UX) of the web application. Create wireframes, prototypes, to visualize the layout. Design database schemas, and system diagrams. Further elaborated in **section 5**.

- **Front-end Development**

Develop the client side of the web application using HTML, CSS, and JavaScript. Implement the UI design, including responsive layouts and interactive elements. Integrate front-end frameworks and libraries as needed. Further elaborated in **section 5.4**.

- **Back-end Development**

Develop the server-side logic, and APIs using C# and .Net framework and implementing user authentication, authorization, and security measures.

- **Database Development**

Design and creating the database schema based on the application's data requirements. Choosing an appropriate database management system and set up the database. Implementing data storage, retrieval, and manipulation operations.

- **Deployment**

Deploy the web application to a production environment, such as a web server, cloud platform, or hosting service.

([www.BrowserStack.com](http://www.BrowserStack.com), 2024)

### **5.2.2 Procedural Programming**

The backend is developed using a robust methodology using C# .net framework. This method is entitled to the procedural code that directly instructs a device on how to finish a task in logical steps. Based on the concept of a procedure call, Procedural Programming divides the program into procedures, which are also known as routines or functions, simply containing a series of steps to be carried out.

- **Predefined functions**

A predefined function is typically an instruction identified by a name. Usually, the predefined functions are built into higher-level programming languages, but they are derived from the library or the registry, rather than the program.



- **Local Variable**

A local variable is a variable that is declared in the main structure of a method and is limited to the local scope it is given. The local variable can only be used in the method it is defined

- **Global Variable**

A global variable is a variable which is declared outside every other function defined in the code. Due to this, global variables can be used in all functions.

- **Modularity**

Modularity is when two dissimilar systems have two different tasks at hand but are grouped together to conclude a larger task first.

- **Parameter Passing**

Parameter Passing is a mechanism used to pass parameters to functions, subroutines or procedures.

*(Sagar Bhatia, 2022)*

### 5.3 Research on Existing ERP's

A comprehensive analysis was undertaken to evaluate the strengths and weaknesses of existing ERP (Enterprise Resource Planning) software. This research aimed to provide a deeper understanding of the landscape, enabling a more informed approach to the ongoing project. The focus was on identifying shortcomings within current solutions to inform strategies for improvement, thereby ensuring that the upcoming work addresses and rectifies these issues, ultimately avoiding the pitfalls observed in existing systems. By learning from both the flaws and virtues of these ERP software solutions, the goal is to develop a more robust and efficient platform that effectively addresses the specific needs and challenges of the intended project.

I made use of my prior experience to ensure that the software meet the needs of users effectively. Drawing from my interactions with users in previous projects, I gained valuable insights into their requirements and preferences. This understanding guided me in designing forms that are intuitive and user-friendly. Additionally, my past experience exposed me to best practices in form design, such as clear labelling, logical layout, and intuitive navigation. I applied these principles to the forms of the ERP software to enhance usability and ensure a positive user experience.

#### **5.3.1 SAGE Group ([www.sage.com](http://www.sage.com), 2019)**

Sage is a globally recognized software company specializing in business management solutions, particularly Enterprise Resource Planning (ERP) and accounting software. Catering to a diverse clientele ranging from small and medium-sized enterprises (SMEs) to large corporations, Sage's products aim to streamline business operations, improve efficiency, and support effective decision-making. *(forbes, 2024)*

Sage offers a glimpse into a versatile suite of tools crafted to address various business needs. Through succinct descriptions, our products are outlined, providing insights into their functionalities and benefits. With a focus on customization, our forms offer flexibility to adapt to diverse business

environments, ensuring easy integration into your workflows. Visual representations, including screenshots and videos, offer a preview of the user interface, allowing you to envision how our forms can enhance your operations. ([www.Sage.com/products](http://www.Sage.com/products), 09/10/2023)

Sage played a crucial role in guiding the development of our software's Receive Order, Purchase Order, Stocks, and Supplier forms through its rich repository of market insights and industry expertise. Sage's comprehensive market research insights provided invaluable guidance regarding prevailing trends, industry standards, and best practices in inventory management, procurement processes, and supplier relationship management. Using this knowledge, the functionalities were tailored of the forms to meet the specific needs of users in this domain, ensuring that the software aligned closely with established industry practices. Furthermore, Sage's functional requirements analysis proved informative in shaping the feature set of our forms. By examining the features offered by Sage and pairing with the unique requirements of our project, we identified essential functionalities that needed integration. This process enabled us to incorporate robust features into our forms, ensuring that they not only met but exceeded the expectations of users.

#### **5.3.1.1 Drawbacks**

- The difficulty of using and exporting Sage's Crystal Reports leads many companies to do custom development for their reporting needs.
- Sage focuses on six industry categories of operations, so many companies need to build custom tools for industries that go beyond those categories.
- Sage X3's core language is a proprietary 4GL language. Few companies have the in-house experts to manage configurations in this language, leading them to seek out consultation and implementation from third-party vendors.

#### **5.3.1.2 Pro's**

- The dashboard interface is set up to aid in both navigation and self-service for troubleshooting, with guided prompts and a "Help" button found on every dashboard screen.
- Sage is built with scalability in mind, whether your business already includes several service lines or subsidiaries, or you plan to grow in the future.
- Sage includes industry-specific modules. If you work in these particular industries, no additional compliance measures or customizations need to be added to the tool

**(Hither, 2021)**

### **5.3.2 Odoo ERP ([www.odoo.com](http://www.odoo.com), 2020)**

Odoo ERP system is enterprise resource planning software used company-wide for the management of business processes. Odoo provides seamlessly integrated functional business apps called Odoo apps that form an ERP solution when combined. Open-source software, Odoo, is available with SaaS subscription pricing as the Enterprise edition or as the Odoo free Community version.

([www.tipalti.com](http://www.tipalti.com), 6/03/2024)

Odoo played a pivotal role in developing the employee page and stock control page. Its HR management module facilitated streamlined employee management, covering onboarding, performance tracking, and payroll administration. Meanwhile, Odoo's inventory management module enabled efficient stock tracking, supplier management, and report generation, enhancing warehouse operations. With Odoo's interface and features, businesses could effectively manage HR and inventory tasks with ease and accuracy.

#### **5.3.2.1 Drawbacks**

- The complexity of Customization: Odoo's customizability is one of its strengths, allowing businesses to tailor the ERP system to their unique needs. However, complex customizations may require expert development skills and could result in increased implementation costs.
- Implementing a new ERP system, such as Odoo, involves a learning curve for employees. They need time to adapt to the new processes and functionalities, which might increase productivity during the transition phase.

#### **5.3.2.1 Pro's**

- Odoo's all-in-one platform streamlines business operations by centralizing data and processes.
- One of Odoo's significant strengths is its customizability. Businesses can tailor the ERP system to their unique workflows and specific requirements.

([www.targetintegration.com](http://www.targetintegration.com), 8/03/2024)

## **5.4 Research on web development**

The concept of web design and development has been around for about as long as websites have existed. It used to have a much simpler definition because website creation used to be a much simpler process. When you compare the first website, which came out in 1991, to modern websites, you can really see how much websites have evolved. Today, creating and maintaining a website is more complex, and involves an entire ecosystem of roles and skill sets. Web design and development is an umbrella term that describes the process of creating a website. Like the name suggests, it involves two major skill sets, web design and web development. Web design determines the look and feel of a website, while web development determines how it functions. Because there isn't always a hard line that separates the two roles, the titles are often used interchangeably. (*InVision, 2018*)

### **5.4.1 Web Designing**

Below are some popular frameworks or technologies used for building web applications:

- **Classic ASP • ASP.NET • ASP.NET Core • ASP.NET Web Forms • ASP.NET MVC**
- **ASP.NET Web Page • ASP.NET API**

***Why ASP webforms was used?***

ASP.NET Web Forms is a framework for building web applications using ASP.NET. It simplifies web development by offering a familiar programming model similar to desktop applications. Key features include server-side controls, event-driven programming, ViewState for preserving control state, and a defined page lifecycle. It's a convenient option for developers transitioning from desktop to web development. It offers a rapid development environment where developers can easily create user interfaces by dragging and dropping server-side controls onto web forms. Its event-driven programming model simplifies handling user interactions and performing server-side processing. Additionally, built-in mechanisms like help manage the state of server-side controls, making it easier to develop stateful web applications. **(Rick-Anderson, 2023)**

## 5.5 Tools and computer languages used

The tools that have been used here to develop the software by not having a cost as I have subscribed to the student version of each which leaves me with limited access but sufficient enough to manage and develop the entire project.

Furthermore, *Visual studio* was opted for frontend design and backend coding as the languages that I pursued to work on were C#, ASP and dot NET framework and SQL server for data management and data injection.

Moreover, establishing a connection between *Visual Studio (Microsoft, 18/11/2023)* and a SQL Server database is a seamless process facilitated by tools like Server Explorer, libraries and Data Connections. Connecting to a SQL Server instance and managing databases can be accomplished visually within the Visual Studio environment by adding the necessary libraries. Additionally, when working on the frontend, particularly in ASP.NET applications, Visual Studio simplifies event handling. For instance, double-clicking on a web control in the Visual Studio Designer generates the corresponding event handler method in the codebehind file (C#), enabling developers to add custom logic effortlessly.

The Primary choice was *Visual Studio Code* which initially proved beneficial for front-end development but posed challenges in integrating C# and .NET. The absence of a dedicated environment for these technologies and the lack of advanced and automated SQL connection features were notable drawbacks. Despite its strengths of installing unlimited extensions, these limitations prompted a reconsideration of the development environment to ensure a more comprehensive solution for diverse project requirements were taken into account.

In *Visual Studio*, the development of files like designer.cs is automated through integrated visual design tools, streamlining the process of creating user interfaces. For instance, in Windows Forms or ASP.NET Web Forms, using the visual designer generates corresponding code in the designer.cs file when controls are added or modified. This automation simplifies the development effort by translating graphical actions into code and maintains a clear separation between generated and custom code for enhanced readability and maintainability. Additionally, Visual Studio automates the creation of other essential files, such as project files and solution files, contributing to a more efficient and consistent development workflow without making it hectic for the programmer or coder.

## 5.6 Data Management and Database

Database and data management are integral aspects of software development, encompassing the systematic organization, storage, retrieval, and manipulation of data. Databases serve as repositories for structured information, and various database management systems (DBMS) cater to diverse needs. Relational databases, like SQL Server, MySQL, and PostgreSQL, use tables and predefined relationships to structure data efficiently. NoSQL databases, such as MongoDB, offer flexibility for unstructured or semi-structured data.

Data management involves designing effective data models, often represented through Entity-Relationship Diagrams (ERDs), to define the structure and relationships within the data. Data Query Languages, like SQL, enable developers to interact with relational databases, allowing for operations such as querying, updating, and managing data. Ensuring data integrity, security, and employing sound modelling practices are crucial for building robust and reliable database systems. ([www.dataentryoutsourced.com](http://www.dataentryoutsourced.com), 2024)

### 5.6.1 Why SQL server

SQL Server is a preferred choice for database management due to its reliability, stability, and robust architecture. Renowned for its scalability, SQL Server accommodates the growth of data and user load, supporting both vertical and horizontal scaling strategies. Security is paramount with features such as encryption, authentication, and authorization, ensuring comprehensive protection against unauthorized access. Particularly advantageous within the Microsoft ecosystem, SQL Server seamlessly integrates with technologies like .NET, Azure, and Visual Studio. ([www.intelequia.com](http://www.intelequia.com), 2021)

Primarily the choice was made based on my prior experience with SQL server as I was not in a position to look forward for a new technology or a database. Given the constraints of time and a preference for mastering existing knowledge over experimenting with new technologies, familiarity with SQL Server ensures a more efficient and reliable development process. Selecting SQL Server as the foundation for the Enterprise Resource Planning (ERP) software is a strategic choice that aligns seamlessly with the vision of a robust and efficient system. Renowned for its reliability and stellar performance, SQL Server ensures the smooth handling of large datasets, accommodating the evolving needs of your expanding business. Additionally, SQL Server's seamless integration capabilities with various Microsoft products enhance interoperability, facilitating a cohesive business environment. Being based on open source it is very easy to access and the vast majority of programmers working in web development have used Microsoft SQL Server.

## 6. Designing of the ERP software

This section is dedicated to exploring the comprehensive integration of web development, database techniques, and design principles within the framework of the software development process. By utilizing these crucial elements, we effectively harness the power

of modern technology to create robust, user-centric solutions that meet the evolving needs of our users.

Web development forms the backbone of our software, serving as the platform through which a dynamic and interactive experience is delivered to the users. Leveraging a combination of front-end and back-end technologies. Database techniques play a vital role in storing, organizing, and retrieving data critical in the functionality of the software.

Through iterative prototyping and user testing, the design choices are refined to match with the preferences and expectations of meeting the desired results, fostering engagement and satisfaction. Additionally, the software process embraces principles of responsive design to ensure consistent user experiences.

## **6.1 Database Design**

A systematic and a basic approach for database design, which is also proposed by Microsoft, entails following a series of key steps. These steps are particularly advantageous for users transitioning from traditional paper-based data storage to digital formats, as well as for modern users navigating migrations between different platforms. Further I'll be listing how this approach meet the requirements of the software developed in terms of database designing: - **(Sukumaran. A; Vergadia. P; Narayanan, Bagirathi, 2023)**

### **6.1.1 Database Importance in ERP**

The primary reason for having a database in this context is to efficiently manage the vast amount of data generated by users interacting with various inventory, stock control, employee management and vendor management forms on the page. Each form captures specific information relevant to the user's needs, and this data needs a structured and organized space to reside. Form as a container, holding valuable pieces of information. The database serves as a centralized storage facility for these containers, ensuring that data from different forms is neatly organized and readily accessible whenever needed. Though, some pages may deal with multiple tables in database simultaneously, each contributing its own set of data to the database. So, essentially, the database plays a crucial role in not only managing individual pieces of data from each form but also in managing the seamless flow and interaction of data from multiple forms across the entire system. Database is the backbone that supports smooth operation of the entire process running, ensuring that users can access the information they need on the software itself which is displayed in the forms of records on the web platform, from the entries stored in the server.

### **6.1.2 Table Design**

Dividing information into tables is a critical step in database design, especially in relational database management systems. Tables organize related data into rows and columns, with each row representing a record and each column representing a data attribute. This process, often guided by normalization techniques, helps eliminate redundancy and maintain data integrity. By structuring data into tables based on the entity-relationship model, databases can efficiently manage and retrieve information according to established relationships between entities.

***(Sukumaran. A; Vergadia. P; Narayanan, Bagirathi, 2023)***

In ERP software tailored for stock control and employee management, the design of tables is intricately tied to the functionality of the forms used for data input and management. Each form corresponds to a specific aspect of data management, such as employee information, inventory items, supplier details, purchase orders, and received orders.

The tables in the database are structured to accommodate the data collected by these forms. For example, one set of tables stores employee-related information, such as names, contact details, salary details, and job roles. Another set of tables manages inventory data, including product names, quantities, prices, and supplier information.

Similarly, tables are designated for storing supplier details, such as company names, contact information, and payment terms. Transactional data associated with procurement and stock receipt processes, like purchase orders and received orders, is stored in separate tables.

By organizing data into tables corresponding to the forms used for data input, ERP software ensures efficient management, organization, and accessibility of information. This structured approach streamlines data entry, retrieval, and analysis, ultimately enhancing operational efficiency and decision-making processes within the organization.

### **6.1.3 Structure of data**

The structure of the data is a crucial factor in deciding the choice and design of a database. There are three widely recognized categories:

- **Structured • Semi-structured • Unstructured**

#### **6.1.3.1 Structured data**

This type of data is typically composed of rows and columns; rows are entities or records and columns are attributes. Structured data is organized in such a way that you can be sure that the data structure will be consistent for the most part throughout the life cycle of that data, except for the possible addition or removal of some attributes altogether. This kind of data is mostly transactional or analytical. ***(Sukumaran. A; Vergadia. P; Narayanan, Bagirathi, 2023)***

In the ERP project, structured data closely aligns with the forms and tables mentioned: Employees, Stocks, Supplier, Purchase Order, Receive Orders, and Sales. Each form represents structured data, with rows representing individual entries or records and columns representing attributes or fields within those records. This consistency ensures that the format and structure of information remain standardized throughout the software's life cycle, facilitating smooth data processing and analysis. The structured data encompasses both transactional and analytical information, covering

daytoday operations like purchase orders and sales transactions, as well as aggregated data used for reporting and decision-making. Overall, structured data in the ERP system ensures systematic organization, efficient data management, and integration across different business processes.

#### **6.1.3.2 Semi-Structured data**

Semi-structured data does not follow a fixed tabular format – that is, a column-row structure. Instead, it stores schema attributes along with data. The attributes for semi-structured data could vary for each record. The major differentiating factor for each kind of semi-structured data is the way they are accessed. ***(Sukumaran. A; Vergadia. P; Narayanan, Bagirathi, 2023)***

Regarding unstructured data, the retrieval of counts for a column in the Stocks form, which is displayed on-screen but not saved in any table or schema, represents an aspect of dynamic data presentation. This unstructured data, while not stored persistently, serves a functional purpose within the software interface, providing users with real-time insights or summaries without the need for permanent storage. It demonstrates how your ERP software effectively utilizes a blend of structured, semi-structured, and unstructured data to cater to various aspects of stock control and employee management. Structured data ensures consistency and reliability, supporting transactional and analytical processes by organizing information into predefined formats. Meanwhile, semi-structured data offers flexibility for dynamic attributes, accommodating variability in certain details across records.

#### **6.1.3.3 Unstructured data**

Unstructured data includes images, audio files, and so on. Unstructured data does not have a definite schema or data model. The amount of unstructured data is much larger than that of structured data. So, the methods by which we store such data are more important than ever. ***(Sukumaran. A; Vergadia. P; Narayanan, Bagirathi, 2023)***. It's pertinent to note that the project does not incorporate such data types. Consequently, while the methods for handling unstructured data remain vital in the broader context of data management, they are not directly applicable within this specific project scope.

#### **6.1.4 Specifying keys**

Each table should include a column or set of columns that uniquely identifies each row stored in the table. This is often a unique identification number, such as an employee ID number or a serial number. In database terminology, this information is called the primary key of the table. Access uses primary key fields to quickly associate data from multiple tables and bring the data together for you. ([www.support.microsoft.com](http://www.support.microsoft.com), 07/03/2024)

The primary keys (PK) in our ERP software are planned to ensure effective data organization and integrity. Each form within the software is designed



with specific considerations for the primary keys, tailored to the nature of the data and the relational structure of the database.

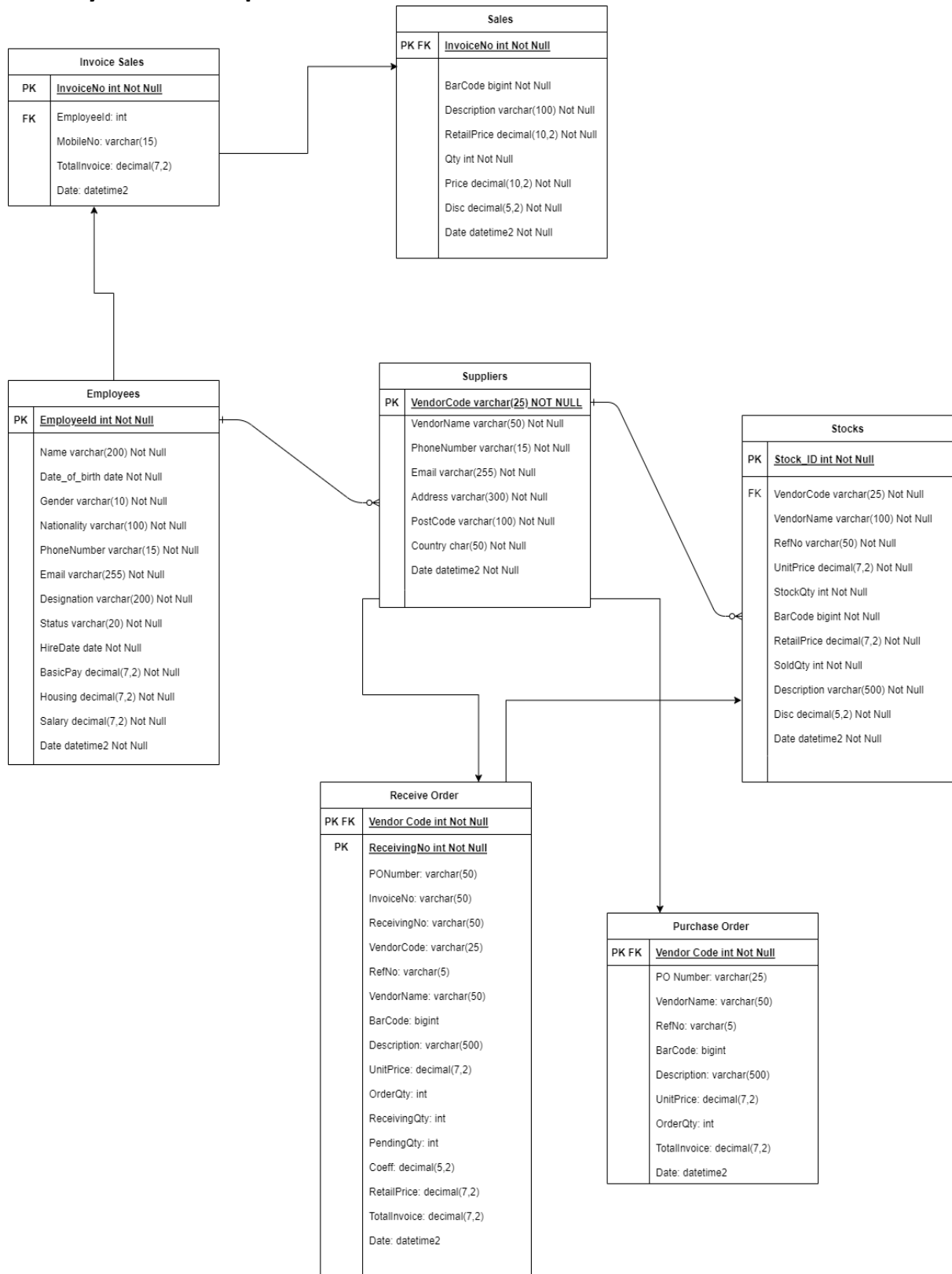
Though, in the Purchase Order form, the PO Number was initially considered as the primary key. However, due to the recurring nature of purchase orders and the need for uniqueness, it was decided that a composite primary key consisting of PO Number and Vendor Code would better serve the purpose. This combination ensures that each purchase order is uniquely identified within the context of a specific vendor.

Similarly, in the Receive Order form, the Receive No was initially contemplated as the primary key. However, given its recurring nature and the potential for duplication, an alternative approach was adopted. Instead, the combination of PO Number, Vendor Code, and Receive No was designated as the composite primary key, providing a more robust and distinct identifier for each received order.

Furthermore, the Employees table employs a straightforward primary key with Employee ID serving as the unique identifier for each employee record. This single attribute primary key simplifies data management and ensures individuality for each employee entry.

In other tables, such as Stocks and Sales, where the primary focus is on inventory and transactional data, no explicit primary key is defined apart from the Suppliers table. In the Suppliers table, Vendor Code serves as the primary key, reflecting its pivotal role as a reference point across various forms and tables within the ERP software. This Vendor Code is often referenced as a foreign key (FK) in other tables, establishing relationships and maintaining data consistency across the system.

## 6.2 Entity-Relationship



## 6.3 User Interface Design

User interface design encompasses the creation of interactive elements that facilitate the interaction between humans and computers. It involves crafting the visual and functional aspects of screens, buttons, forms, and navigation elements within an application. These

elements include input controls like checkboxes and dropdown lists, navigational elements such as search fields and sliders, informational components like notifications and progress bars, and containers for organizing content effectively. UI design focuses on interactivity, visual representation, and information architecture to ensure interfaces are intuitive and visually appealing. It involves considerations such as how elements behave upon interaction and the overall look of the interface, including colour schemes, typography, and iconography. While UI design addresses the tangible aspects of the interface, user experience (UX) design encompasses the broader journey of users interacting with the product. UX design involves understanding user needs, conducting research, and optimizing the overall experience to meet those needs. While UI design focuses on the surface-level appearance and functionality of the interface, UX design delves deeper into understanding user behaviours and emotions to create seamless and satisfying experiences. Together, UI and UX design collaborate to create interfaces that are both visually appealing and userfriendly, ultimately enhancing the overall usability and enjoyment of the product. **(Stevens E. 2022).**

Upon researching I came across an article from a book which described the golden rule for User interface design and was written by Ben Shneiderman an American computer scientist, a Distinguished University Professor in the University of Maryland Department of Computer Science, and the founding director (1983-2000) of the University of Maryland HumanComputer Interaction Lab. He conducted fundamental research in the field of human– computer interaction, developing new ideas, methods, and tools such as the direct manipulation interface, and his rules of design. **(Shneiderman. B, 2016)**

#### **6.3.1 Build User Interface consistent**

Consistent UI means keeping things the same across the board. That means using the same kinds of designs, using the same words for instructions, making all the menus and screens look alike, and using the same commands everywhere you can. When you do this, it makes the whole app feel smoother and easier to use because everything works in the same way. It's like having a familiar road to travel on – you know what to expect and where everything is. This consistency helps users feel more comfortable and confident while using the app. **(Shneiderman, B. and Plaisant., C, 2010)**

The similarities between the rule of consistent UI and the aspects of the ERP software's UI design are clearly displayed. Just as the rule emphasizes keeping things uniform across the entire application, the software maintains consistency in various aspects. In terms of design, the software ensures uniformity by employing identical design elements across all forms used for data insertion, updates, and deletions, as well as for pages dedicated to viewing data. The layout and structure of these forms and pages remain largely consistent, with the only discernible difference being the top heading, which serves as the form's title. Additionally, screens and menus exhibit a consistent colour scheme throughout, contributing to a cohesive visual experience. This consistency aids users in effortlessly navigating the application, as they encounter familiar visual cues across different sections. Regarding language, the rule suggests using identical words for instructions. Similarly, the software maintains

consistency by employing similar terminology and labelling for the various types of errors seen while working on it across different forms and screens.

### **6.3.2 Allow users to navigate easily via shortcuts**

Allowing users to navigate easily via shortcuts refers to the provision of alternative, often keyboard-based, methods for users to quickly access and interact with various features and functions within a software application or system. These shortcuts typically involve combinations of keys or sequences that perform specific actions, such as navigating to different sections of the application, executing commands, or performing common tasks. **(Shneiderman, B. and Plaisant. C, 2010)** In the ERP software, users can perform basic shortcut functions to simplify the workflow and enhance efficiency. One of the fundamental shortcuts available is the use of keyboard commands for common actions such as copy, paste, and saving data. Though, users can easily copy data from one field to another by simply selecting the desired text and pressing "Ctrl + C" to copy and "Ctrl + V" to paste. This functionality mirrors standard keyboard shortcuts used in most applications, allowing users to perform these actions seamlessly without relying on context menus or toolbar buttons.

Furthermore, when entering or inserting data in the ERP system, users can pursue by pressing "Enter" to save the information instead of clicking the designated "Save" button. This feature significantly reduces the time and effort required to commit changes, as users can easily move from one data entry field to the next and finalize their inputs.

Similarly, for deleting records in the ERP software, users can utilize a keyboard shortcut, such as "delete", to trigger the deletion process. Upon entering the desired records unique ids and pressing the designated shortcut, the software prompts the user to confirm the deletion, ensuring that critical data is not accidentally removed. By providing this shortcut, users can efficiently manage data without the need to navigate to the delete button.

### **6.3.3 Design dialog to yield closure**

Sequences of actions should be organized into groups with a beginning, middle, and end. Informative feedback at the completion of a group of actions gives users the satisfaction of accomplishment, a sense of relief, a signal to drop contingency plans from their minds, and an indicator to prepare for the next group of actions.

**(Shneiderman. B, 2010)**

Moreover, in the ERP software, messages such as "employee added or deleted", "purchase complete," "order received," "record updated," and "record deleted" serve as informative feedback to users, indicating the completion of specific actions within the system. These messages prompt closure for users by providing confirmation and reassurance that their intended tasks have been successfully executed. This immediate feedback acknowledges the user's action and confirms that the transaction has been processed successfully. This feedback not only gives users a sense of accomplishment but also assures them that their requests have been fulfilled. Similarly, when a user updates or deletes a record, the software generates

messages such as "record updated" or "record deleted" to indicate the completion of these actions. This feedback informs users that their changes have been successfully applied, allowing them to proceed with confidence knowing that the system has registered their updates or deletions.

#### **6.3.4 Prevent error as much as possible**

Make the UI as easy to use as possible by preventing serious user errors. So, from greying out menu items that are not available to preventing users from typing in alphabets in fields that ask for phone numbers, try to prevent error as much as possible. However, occasionally errors happen. Though if a user has made an error, make sure to offer them a clear explanation to understand the error and an easy solution to solve it. **(Shneiderman. B, 2010)**

The backend C# code implements several strategies to minimize user errors and ensure a smooth user experience. Through a combination of if-else conditions and try-catch blocks, it guards against potential issues and provides clear feedback to users when errors occur.

Though, the code employs validation checks using if-else conditions to ensure that critical fields are not left empty. It verifies that essentials are all filled out before proceeding with database operations. This prevents incomplete submissions and potential data inconsistencies.

Although, the code incorporates error handling mechanisms using try-catch blocks to anticipate and manage exceptions gracefully. For example, when executing database queries or commands, try-catch blocks capture potential errors, such as database connection failures or SQL query execution issues. By handling these exceptions, the code prevents abrupt crashes and provides users with meaningful error messages, facilitating better understanding and resolution of issues.

Furthermore, the code implements specific checks to prevent common user errors. For example, it validates email addresses to ensure they contain "@" and "." symbols, helping users avoid entering invalid email formats. Similarly, it verifies phone numbers to ensure they consist only of digits or allowed symbols like "+", preventing users from inputting alphabetic characters or other invalid characters. The code includes clear comments to explain each block's purpose, behaviour, and any special conditions. These comments help developers understand the error handling logic easily.

#### **6.3.5 Minimize memory load**

A key rule for making user interfaces easy for people to use is minimizing cognitive load. Cognitive load (or memory load) can reduce a user's capacity to perform important tasks, so it's critical that computers take over the burden of memory from them as much as they can. **(Shneiderman, B. and Plaisant. C, 2010)**

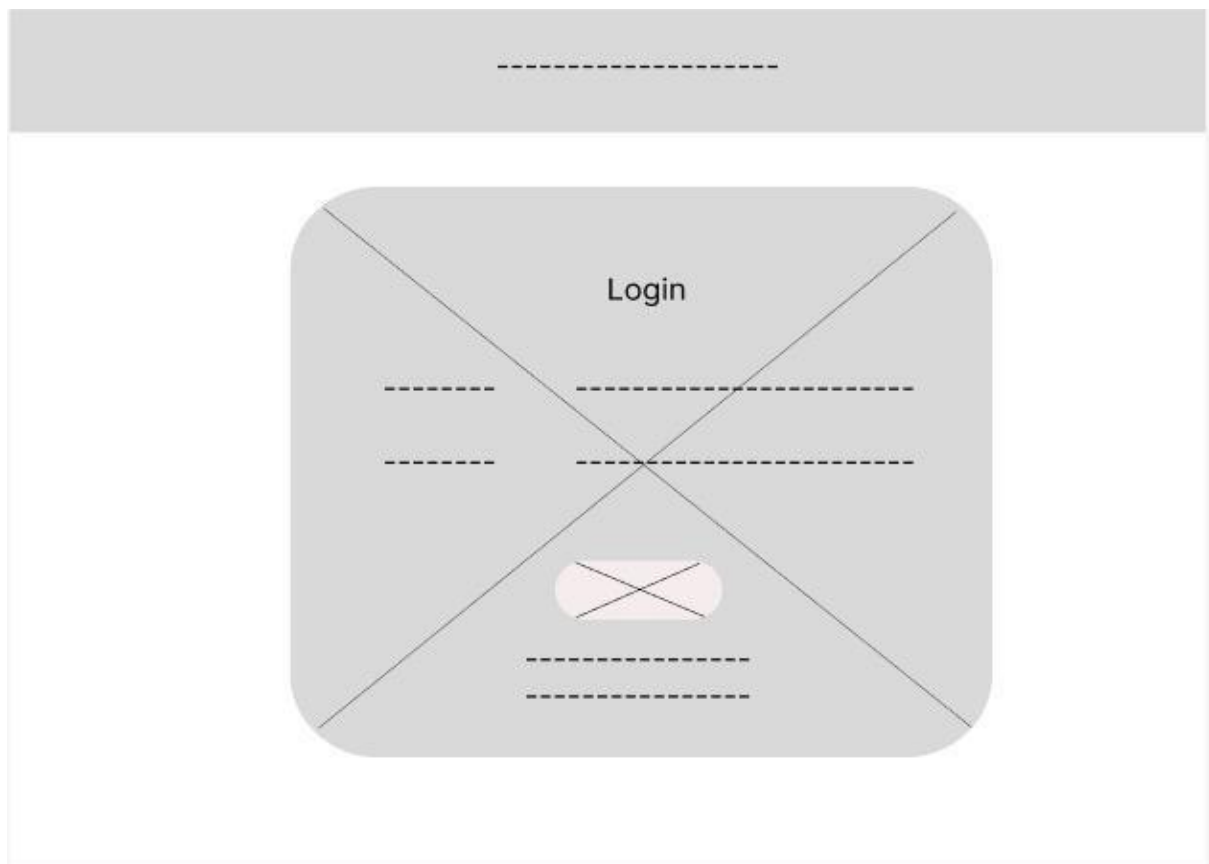
Through the code, implementation of few strategies was taken place to minimize memory load by requiring minimal input from users, particularly when retrieving data. Each form requires users to provide a unique identifier, such as a PO Number, Receive No, Employee No Vendor Code, Invoice Number, to retrieve specific records from the database for the particular form. By asking for only this essential piece of

information, users are relieved of the burden of recalling detailed data or navigating through extensive menus.

Furthermore, development of a separate page "View Data" for each form, where users can conveniently access records using predefined unique IDs made accessing the data more convenient. This approach further reduces the load by providing users with a straightforward interface to retrieve data without the need to remember complex queries or navigate through multiple screens.

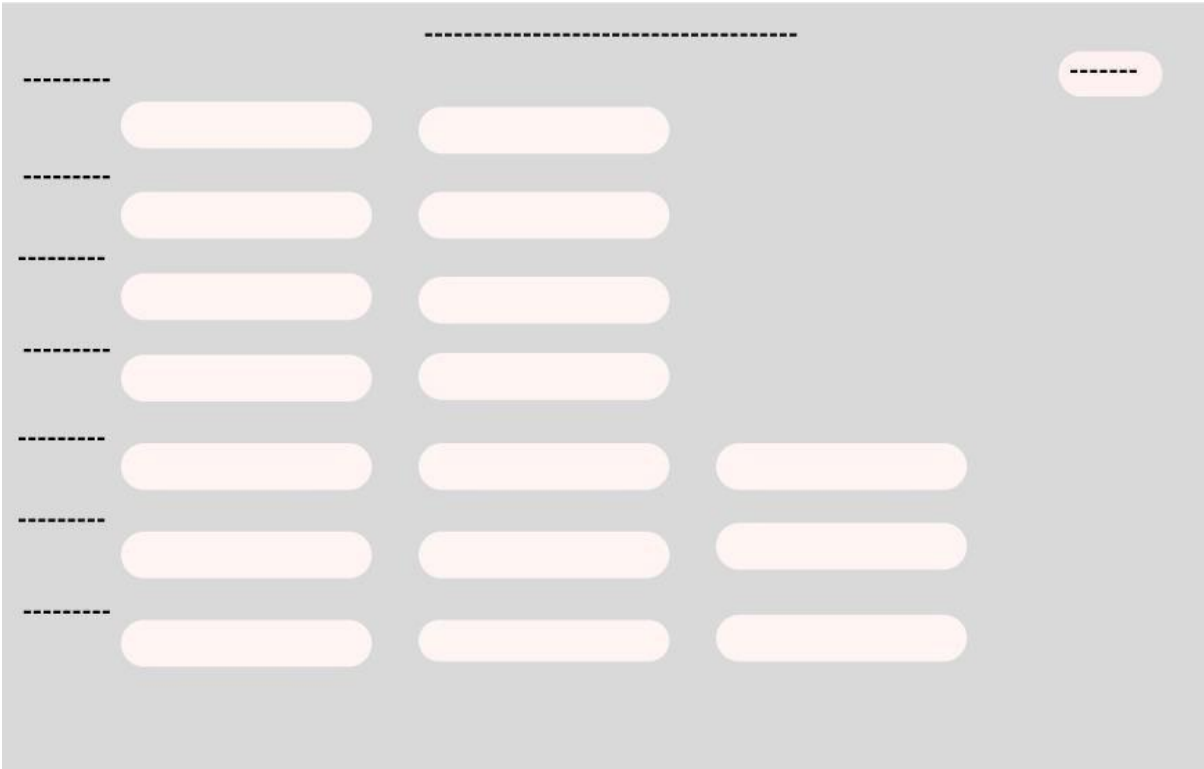
## 6.4 Prototype and Wireframes

### 6.4.1 The Login Page





**6.4.2 HomePage**



## The ERP Software.

[Log Out](#)

|           |                                |                               |                                |
|-----------|--------------------------------|-------------------------------|--------------------------------|
| Employees | <a href="#">Employees Form</a> | <a href="#">EMPLOYEE View</a> |                                |
| Stocks    | <a href="#">Stocks Form</a>    | <a href="#">Stocks View</a>   |                                |
| Supplier  | <a href="#">Supplier Form</a>  | <a href="#">Supplier View</a> |                                |
| Purchase  | <a href="#">Purchase Form</a>  | <a href="#">Purchase View</a> |                                |
| Receive   | <a href="#">Receive Form</a>   | <a href="#">Receive View</a>  | <a href="#">Receive Report</a> |
| Sales     | <a href="#">Sales Form</a>     | <a href="#">Sales View</a>    | <a href="#">Sales Report</a>   |

### 6.4.3 Forms Page

## ERP SOFTWARE

ERP Things!

[Home](#)
[Tools](#)
[Services](#)
[Contact](#)
[Logout](#)

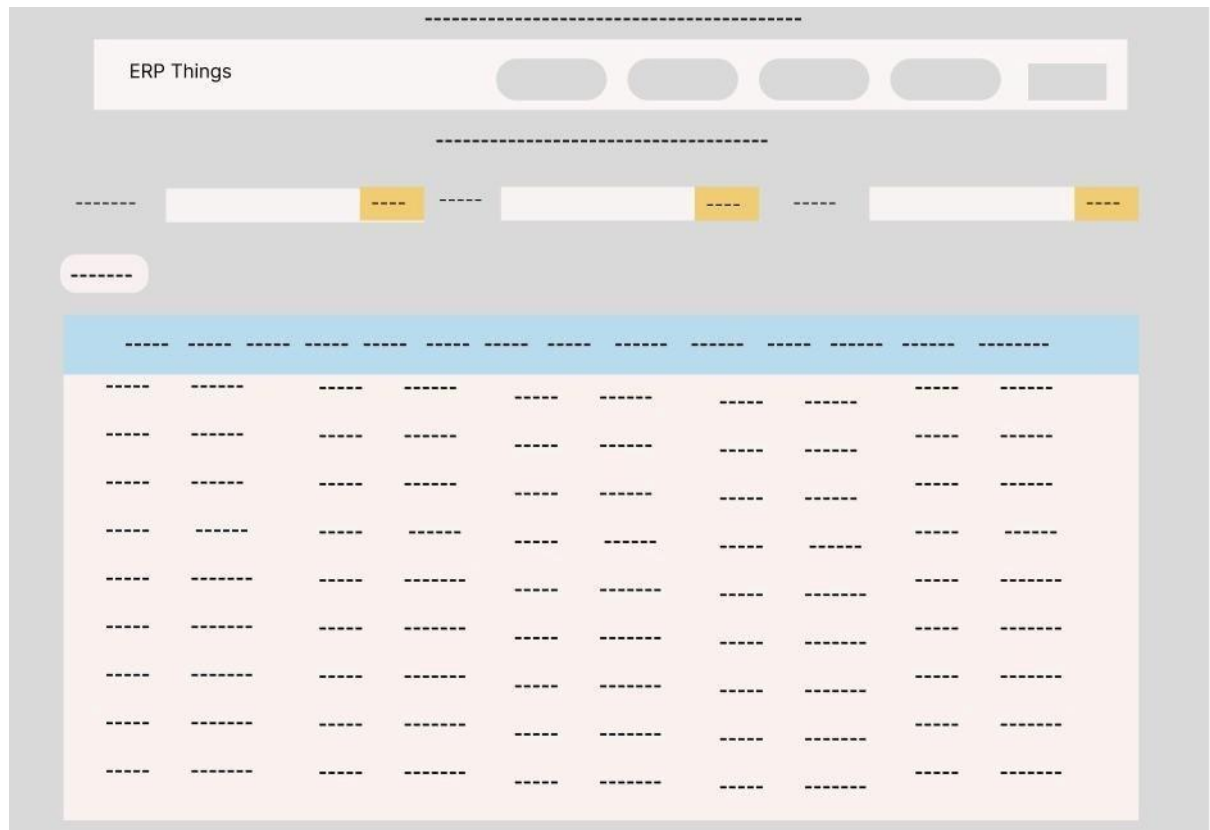
### Purchase Goods

|              |                                            |              |                      |              |                      |
|--------------|--------------------------------------------|--------------|----------------------|--------------|----------------------|
| PO. No.:     | <input type="text"/>                       | Vendor Code: | <input type="text"/> | Vendor Name: | <input type="text"/> |
| Ref.no:*     | <input type="text"/>                       | Category:    | <input type="text"/> |              |                      |
| BarCode:     | <input type="text"/>                       |              |                      |              |                      |
| Unit Price:  | <input type="text"/>                       |              |                      |              |                      |
| Order Qty*:  | <input type="text"/>                       |              |                      |              |                      |
| Total Price: | <input type="text"/>                       |              |                      |              |                      |
| Description: | <input style="height: 40px;" type="text"/> |              |                      |              |                      |

[Edit](#)
[Update](#)
[Delete](#)
[Save](#)



#### **6.4.4 View Page Wire frame**



### **6.5 General Design Issues**

During the development process, several general design issues were encountered, each posing unique challenges that required careful consideration and strategic planning.

#### **6.5.1 Security**

Security emerged as another critical design concern, given the sensitive nature of the data being managed by the software. Protecting against unauthorized access, data breaches, and other security threats necessitated the implementation of robust security measures at various levels of the application stack. This involved employing encryption techniques, access control mechanisms, and thorough security audits to mitigate potential vulnerabilities and safeguard user data.

To address security concerns, a robust login page was implemented as the first line of defence against unauthorized access. This login page required users to authenticate themselves with valid credentials before gaining access to the system. Additionally, encryption techniques were applied to sensitive data stored within the database to prevent unauthorized viewing or tampering.

#### **6.5.2 Performance**

Performance emerged as a critical design concern during the development process, as the software needed to efficiently handle large volumes of data and user interactions while maintaining responsive and smooth operation. Ensuring optimal performance required careful optimization of code, database queries, and system architecture to minimize latency and maximize throughput.

For maintainability, Code was optimized by reducing unnecessary operations, while database queries were fine-tuned with indexes and caching mechanisms to minimize latency and improve response times. These measures ensured the ERP software operated efficiently, even under heavy loads.

### **6.5.3 UI/UX**

Significant attention was also directed towards user experience (UX) design within the software development process. The focus was on crafting intuitive interfaces, refining user workflows, and integrating user feedback to ensure high levels of user satisfaction and usability. By emphasizing UX design principles like simplicity, consistency, and accessibility, the software aimed to provide engaging and userfriendly experiences tailored to the needs and expectations of its target audience.

While prioritizing functionality over UI/UX design, the software was developed with a strong emphasis on user experience. Intuitive interfaces were crafted to seamlessly guide users through the system, streamlining workflows and reducing the complexity of common tasks. This approach aimed to enhance efficiency and usability for all users interacting with the software.

## **7. Implementation**

The preceding sections detail the rationale behind the development of the software, including planning strategies and conceptual frameworks. Moving forward, the focus shifts towards showcasing the tangible progress made during the development process. This will encompass configuring the Integrated Development Environments (IDEs), setting up the database infrastructure, leveraging gathered information, and elucidating the practical application of the languages employed in the project.

### **7.1 System setup**

Setting up a development environment in software development involves configuring the software and hardware on which code will be written, tested, and executed. A wellconfigured environment ensures consistency across different development stages and among team members, leading to fewer bugs and deployment issues. This typically involves tasks such as installing and configuring operating systems, setting up hardware components, configuring software applications, establishing network connections, and ensuring compatibility and security measures are in place. ([www.reintech.io](http://www.reintech.io), 01/01/2024)

### **7.2 IDE (Integrated Development Environment) Setup**

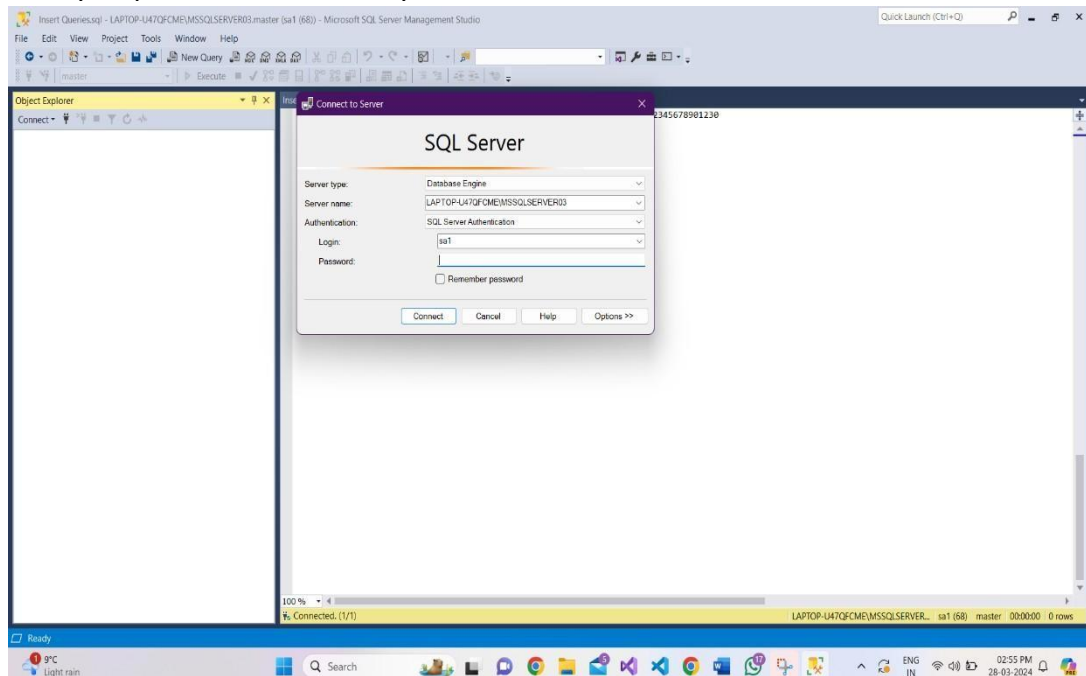
As indicated above, Visual Studio served as the primary integrated development environment (IDE) for crafting the ERP software. **Section 4.4** elaborates on the alignment facilitated by this tool between the backend (C#) and frontend (aspx) languages, highlighting its exceptional convenience in the development process.

### 7.2.1 Downloading and Setting up Visual Studio

Visual Studio 2019 Community Edition (**Microsoft, 2019**) was chosen for its accessibility, as it provides a comprehensive set of features suitable for individual developers, open-source projects, academic research, education, and small professional teams. Additionally, it's free to use, making it an attractive option for those starting with software development. Moreover, installation process was exactly followed guided by the instruction and the platform was initialized. ASP.NET have been chosen as a workload during the Visual Studio installation process for its suitability in web development. ASP.NET provides a robust framework for building dynamic web applications and services, making it a popular choice for developing ERP systems that require web-based interfaces and functionality. The final step was naming of the project which was assigned as **ERP Project** and was saved on my desktop.

### 7.2.2 Installing and configuring SSMS (SQL server Management Studio)

In meeting the security criteria, the decision was made to implement SQL Server authentication, requiring the creation of usernames and passwords. However, this posed a challenge as I lacked prior experience in setting up credentials for server authentication. To overcome this obstacle, I turned to a YouTube video tutorial by Code by Salman. Through this resource, I gained valuable insights into the process of enabling usernames and passwords for SQL authentication, thus addressing the security requirements effectively.



## 7.3 Database operations

### 7.3.1 Employees Entity:

- In the ERP software, the Employees entity is responsible for managing employee-related information. It aligns perfectly with the software's objective of facilitating employee management within the organization.
- Attributes such as EmployeeId, Name, Date\_of\_birth, and Gender are crucial for accurately identifying and categorizing employees within the system. This aligns with the ERP software's need to maintain detailed employee records for various HR processes.
- Other attributes like PhoneNumber, Email, Designation, and Status contribute to effective communication, role assignment, and status tracking of employees, ensuring seamless workflow management.
- The inclusion of attributes like BasicPay, Housing, and Salary enables the ERP software to manage payroll and compensation-related tasks efficiently, aligning with the software's focus on comprehensive employee management.

#### **7.3.2 Suppliers Entity:**

- The Suppliers entity plays a pivotal role by centralizing information about vendors and suppliers. It aligns perfectly with the software's objective of managing external relationships and procurement processes.
- Attributes such as VendorCode, VendorName, PhoneNumber, and Email are essential for identifying and contacting suppliers, enabling streamlined communication and collaboration.
- Other attributes like Address, PostCode, and Country provide valuable location details, facilitating logistics and supply chain management within the ERP system.
- The inclusion of the Date attribute ensures that the ERP software maintains accurate records of supplier interactions, order placements, and deliveries, aligning with the software's emphasis on tracking transactional data.

#### **7.3.3 Stocks Entity:**

- In the software, the Stocks entity is integral to functionality, enabling efficient management of inventory and stock control processes.
- Attributes such as VendorCode, RefNo, and BarCode allow the ERP system to uniquely identify and categorize stock items, facilitating accurate tracking and management of inventory levels.
- Key attributes like UnitPrice, StockQty, and RetailPrice provide essential pricing and quantity information, enabling the software to monitor stock levels, pricing trends, and profitability effectively.
- The inclusion of attributes like Description and Disc enhances the ERP software's capability to manage diverse product catalogs and pricing strategies, aligning with the software's focus on supporting various business models and product offerings.

#### **7.3.4 Purchase Order Entity:**

- Furthermore, the PurchaseOrder entity serves a critical role by enabling the management of procurement processes and purchase transactions.
- Attributes such as PONumber, VendorCode, and RefNo allow the ERP system to uniquely identify and categorize purchase orders, facilitating accurate order tracking and fulfillment.
- Key attributes like UnitPrice, OrderQty, and TotalInvoice provide essential details about purchase transactions, enabling the software to generate accurate invoices, monitor spending, and manage budgets.
- The inclusion of attributes like Description and Date ensures that the ERP software maintains comprehensive records of purchase transactions, including item details and transaction timestamps, aligning with the software's emphasis on audit trails and transaction history.
- By capturing essential information such as VendorName and Date, the ERP system can streamline vendor management processes, track supplier performance, and maintain transparency in procurement operations.

#### **7.3.5 Receive Order Entity:**

- The ReceiveOrder entity plays a crucial role in managing the receipt and processing of goods from suppliers, ensuring accurate inventory management and stock reconciliation.
- Attributes such as PONumber, InvoiceNo, and ReceivingNo enable the ERP system to link received goods to specific purchase orders and invoices, facilitating seamless order fulfillment and payment reconciliation processes.
- Key attributes like ReceivingQty, PendingQty, and TotalInvoice provide essential details about received goods, enabling the software to update inventory levels, calculate pending quantities, and generate accurate financial reports.
- The inclusion of attributes like RetailPrice and Coeff enhances the ERP software's capability to calculate pricing and profitability metrics, supporting decision-making processes related to pricing strategies and product margins.

#### **7.3.6 Sales Entity:**

- In the database of the ERP software, the Sales entity serves as a central repository for managing sales transactions and customer interactions, supporting revenue generation and customer relationship management.
- Attributes such as InvoiceNo, BarCode, and Description allow the ERP system to track sales orders and product details, enabling accurate order fulfilment and inventory management.
- Key attributes like RetailPrice, Qty, and Price provide essential information about sales transactions, enabling the software to calculate total sales revenue, monitor sales performance, and generate financial reports.
- The inclusion of attributes like Disc and Date enhances the ERP software's capability to apply discounts and promotions, track sales trends, and analyse customer buying behaviour, supporting marketing and sales strategies.

### **7.3.7 Invoice Sales Entity:**

- In the software, the InvoiceSale entity facilitates the generation and management of sales invoices, ensuring accurate billing and revenue recognition processes.
- Attributes such as EmployeeId, MobileNo, and TotalInvoice allow the ERP system to associate sales transactions with specific employees and customers, enabling personalized service delivery and customer billing.
- The inclusion of attributes like Date ensures that the ERP software maintains accurate records of invoice generation and payment timelines, supporting financial reporting and compliance requirements.
- By capturing essential information such as InvoiceNo and TotalInvoice, the ERP system can streamline invoice management processes, track outstanding payments, and reconcile financial accounts effectively.

## **7.4 Front-End Design**

### **7.4.1 Input Forms**

The ASPX framework was utilized to create interactive input forms tailored to the specific requirements of each entity within the ERP software. For instance, Employee.aspx facilitated comprehensive employee management by strategically positioning text boxes and labels for efficient data entry and retrieval. Similarly, Stocks.aspx, Suppliers.aspx, Sales.aspx, PurchaseForm.aspx, and ReceiveForm.aspx employed ASPX controls to construct input fields for essential attributes like VendorCode and RefNo, streamlining inventory management and purchase order processes.

CSS played a crucial role in styling elements such as text boxes, divs, and tables, ensuring a balanced and engaging interface. Background colors were selected to align with software requirements and create an appealing visual identity, enhancing organization recognition. Font sizes were optimized for readability across different screen sizes, and navigation elements were enhanced with CSS transitions for smoother user interaction.

Bootstrap utilities complemented ASPX and CSS implementations by providing handling for layout, typography, and interactive elements. This facilitated the quick customization of navigation bars and ensured a consistent and modern look across all forms and pages. Additionally, Bootstrap's responsive grid system enhanced accessibility and usability across various devices, contributing to a seamless user experience within the ERP software.

#### **7.4.1.1 Navigation Bar Implemented**

A navigation bar was implemented on each form to facilitate easy navigation between different forms, view data pages and to the report creation page of the ERP software. This navigation bar, created using ASPX, Bootstrap and CSS, provided users with intuitive access to various functionalities, including switching between input forms and accessing other modules of the application. Additionally, the navigation bar featured a prominently placed logout button, ensuring convenient access for users to securely sign out of their sessions when needed.

#### **7.4.1.2 Buttons on Each Form**

Each input forms within the ERP software was thoughtfully designed to include a variety of buttons, each serving a specific purpose to facilitate user interaction and data manipulation. In addition to commonly used buttons like "Save," "Edit," and "Delete," which were essential for actions such as saving, editing, and deleting data, several other buttons were placed throughout the forms.

These additional buttons were responsible for performing various tasks, such as carrying out calculations, generating new IDs, and enabling or disabling primary buttons based on certain conditions or user inputs. Their inclusion enhanced the functionality and flexibility of the forms, allowing users to perform a wider range of actions efficiently.

#### **7.4.1.3 Purpose of the Forms**

The purpose of these input forms varied based on the entity they represented within the ERP software. Generally, the forms were designed for data insertion, retrieval, and management. For example, the Employee form allowed users to input and update employee information, while the Stocks form facilitated inventory management by capturing stock-related data. Similarly, the Suppliers form enabled the recording and tracking of vendor details, while the Purchase and Receive forms facilitated purchase order creation and goods receipt processes, respectively.

The screenshot displays a web browser window with the URL <https://localhost:44324/ReceiveForm>. The application title is "THE ERP SOFTWARE". The navigation bar includes "ERP Things", "Home", "Tools", "Contact", "Performance", and a red "Logout" button. The main heading is "Receive Order".

The form is organized into three columns of input fields:

- Left Column:** PO Number, Invoice No., Receiving No., Vendor Code, Ref No. (with a blue "Get" button), Bar Code, Vendor Name, and Description.
- Middle Column:** Order Qty, Receive Qty, and Pending Qty.
- Right Column:** Unit Price, Coeff. (with a yellow "Calc" button), Retail Price, and Total Invoice.

At the top right, there is a "Received PO: 0" field. At the bottom right, there are three buttons: "Receive Order", "Delete Record", and "Clear". The Windows taskbar at the bottom shows the date as 28-03-2024 and time as 02:57 PM.

#### **7.4.2 View Data Page**

In the ERP software, all view pages, including `EmployeeView.aspx`, `StocksView.aspx`, `SuppliersView.aspx`, `PurchaseView.aspx`, and `ReceiveView.aspx`, are equipped with a GridView control to efficiently present data in a structured tabular format. This approach is highly advantageous as it allows users to easily visualize and interact with large datasets. The

GridView control is a built-in feature provided by ASP.NET in Visual Studio, simplifying the process of data binding and displays.

To assign the GridView control, developers typically utilize ASPX markup to define the control within the view page. Through ASP.NET's data-binding capabilities, the GridView is bound to a data source, such as a SQL Server database, enabling it to dynamically populate rows and columns with retrieved data. The data is fetched using server-side logic, often implemented in C# code-behind files, and then bound to the GridView for display.

#### **7.4.2.1 Purpose of the Pages**

The primary purpose of the view pages is to provide users with a structured presentation of data. Users can easily visualize, filter, and interact with the displayed data using the search and get buttons. The layout of buttons and textboxes within HTML tables ensures proper alignment and presentation, enhancing the user experience by providing a clear visual association between the buttons and the corresponding data fields.

#### **7.4.2.2 Buttons on View Page**

Additionally, the view pages feature search and get buttons to facilitate data retrieval and filtering. These buttons, implemented using ASPX Button controls, allow users to search for specific records or retrieve detailed information based on predefined criteria. The search functionality enhances user productivity by enabling quick access to relevant data within the displayed dataset. The button present along with the textbox retrieves the data based on the data inputted in the textbox whereas the search button derives data based on the combination of data from all the textboxes.

Regarding the layout, buttons and associated textboxes on view pages are often organized within HTML tables to ensure proper alignment and presentation. Each button is typically coupled with relevant textboxes or data fields, grouped together within a row of the GridView. This layout enhances user experience by providing a clear visual association between the button and the corresponding data it acts upon.

#### **7.4.2.3 Navigation Bar**

Furthermore, the navigation bar present on view pages mirrors the design of the navigation bar on input forms. This consistency in design, implemented using ASPX, CSS, and possibly Bootstrap for styling and responsiveness, ensures ease of use and familiarity for users. By maintaining a uniform navigation interface across different sections of the ERP software, users can effortlessly identify and access the desired functionality, enhancing overall usability and user satisfaction.



| PONumber | InvoiceNo    | ReceivingNo | VendorCode | RefNo | BarCode       | Description           | UnitPrice | OrderQty | ReceivingQty | PendingQty | Coeff | RetailPrice | TotalInvoice | Date       |
|----------|--------------|-------------|------------|-------|---------------|-----------------------|-----------|----------|--------------|------------|-------|-------------|--------------|------------|
| PO1890   | 7755221368   | R2233556    | V2         | 005   | 5678901234567 | Software Licenses     | 12.99     | 50       | 35           | 15         | 3.02  | 39.23       | 454.65       | 26/01/2024 |
| PO1890   | IN77898569   | 2656256545  | V2         | 005   | 5678901234567 | Software Licenses     | 12.99     | 50       | 10           | 5          | 3.02  | 39.23       | 129.90       | 26/01/2024 |
| PO1890   | 7755221785   | R223355645  | V2         | 005   | 5678901234567 | Software Licenses     | 12.99     | 50       | 5            | 0          | 3.02  | 39.23       | 64.95        | 26/01/2024 |
| PO1894   | 778946622    | 222336488   | 789456     | 357   | 555666223322  | Kings Heat            | 50.35     | 45       | 35           | 10         | 3.02  | 152.06      | 1762.25      | 30/01/2024 |
| PO1891   | 545154544552 | 3245465262  | 51852      | 010   | 1234567890123 | Semiconductor Devices | 37.75     | 10       | 7            | 3          | 3.02  | 114.01      | 264.25       | 30/01/2024 |
| PO1891   | 7789666      | 222566      | 51852      | 010   | 1234567890123 | Semiconductor Devices | 37.75     | 10       | 3            | 0          | 3.02  | 114.01      | 113.25       | 30/01/2024 |
| PO1894   | 7733102      | 56230...    | 789456     | 357   | 555666223322  | Kings Heat            | 50.35     | 45       | 8            | 2          | 3.02  | 152.06      | 402.80       | 30/01/2024 |
| PO1895   | 78555220     | 1003200     | V2         | 020   | 741594651656  | blue Can              | 1.00      | 100      | 90           | 10         | 3.02  | 3.02        | 90.00        | 05/02/2024 |

### 7.4.3 Report formulation page

The ERP software represents a page dedicated to generating reports, specifically for Purchase Orders, Receive Orders and Stocks. This page, is named accordingly to the relation with its form or view page. Moreover, it utilizes ASP.NET technologies such as ASPX for markup and C# for server-side logic. It incorporates various HTML elements and controls to facilitate the generation of purchase-related reports.

#### 7.4.3.1 Purpose of the page

The primary purpose of the Report page is to provide users with the functionality to generate detailed reports related to various transactions within the ERP software, including purchase orders, receive orders, and stocks. Upon clicking the "Generate" button, the system retrieves relevant data from the database based on the specified criteria and generates a comprehensive report.

This report typically includes details such as PO number or receiving number, Vendor details, Ref number, BarCode, Description, Unit Price, Order Quantity, and Total Amount for each transaction, whether it's a purchase order, receive order, or stock record. The generated report serves as a valuable tool for analysing transactional data, tracking expenses, monitoring inventory levels, and making informed business decisions within the organization. By offering flexible filtering options and comprehensive reporting capabilities, the Purchase Report page enhances operational efficiency and facilitates datadriven decision-making processes across various aspects of the ERP system.

#### 7.4.3.2 Buttons on Report Page

The Report page features a single button named "Generate" represented by the <asp:Button> control. This button triggers the generation of the report when clicked. It is styled using CSS classes to ensure visual consistency with the overall design of the page. Upon clicking the "Generate" button,

serverside logic defined in the code-behind file PurchaseReport.aspx.cs is executed to retrieve relevant data and generate the report.

#### **7.4.4 Home Page and Login Page**

The Home Page and Login Page serve as main components of the ERP software's frontend interface, each playing a distinct role in user interaction, authentication, security and navigation.

##### **7.4.4.1 Login Page (Index.aspx):**

The Login Page, designated as Index.aspx, serves as the entry point for users to access the ERP software. The choice of naming it as "Index.aspx" aligns with conventional web development practices, where the "index" denotes the main or starting page of an application.

HTML elements are strategically utilized to structure the login page, enhancing the layout and positioning of essential components. Divisions and tables are employed to organize elements such as textboxes for username and password inputs, error message display, and login button.

To ensure a simple user experience, Bootstrap framework is integrated into the login page. Bootstrap's classes are applied to textboxes, buttons, and links, providing consistent styling and responsiveness across different devices and browsers.

Textboxes for username and password inputs are further customized using CSS, with the type attribute of the username textbox set to "email" for browser-based email validation. This enhances data integrity and user experience by facilitating correct input formats.

The login button is styled using Bootstrap and CSS, enhancing its appearance and interactivity. Custom CSS classes and Bootstrap's predefined button styles are applied to create visually appealing buttons with hover effects for improved usability.

Links for "Create New Login" and "Forgot Password" are also customized using CSS to provide visual cues for users. CSS styles such as text-decoration and colour are applied to ensure these links stand out on the page and guide users to relevant actions.

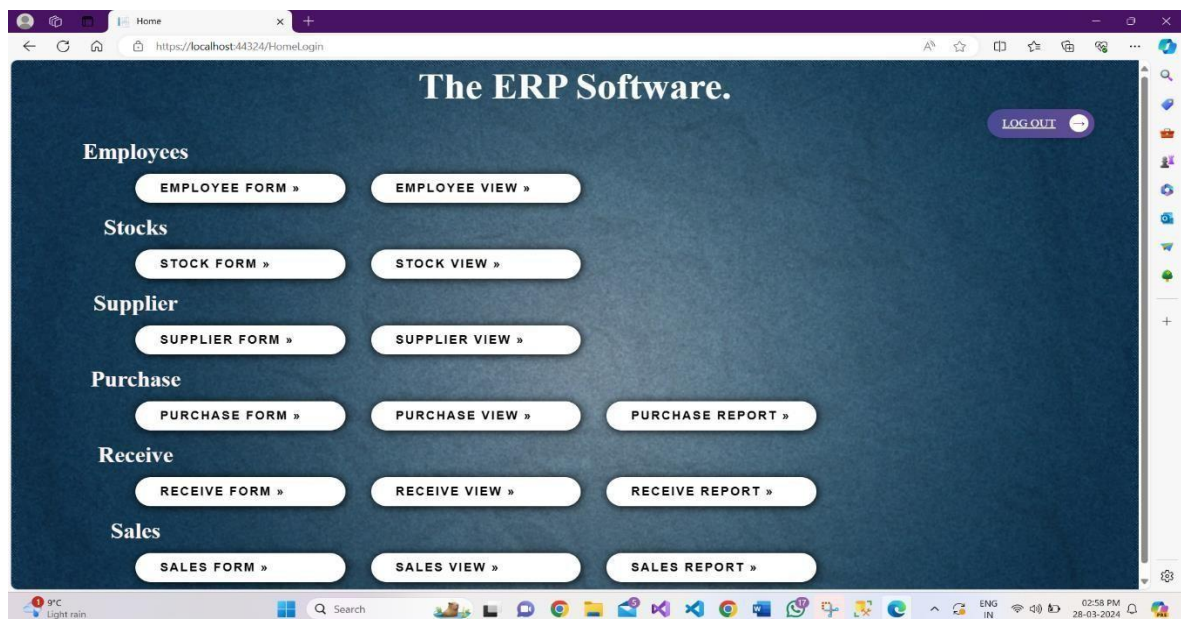
Upon successful login, users are redirected to the Homepage (HomeLogin.aspx), the central hub for accessing different modules and functionalities within the ERP software. Conversely, clicking on the "Logout" button redirects users back to the Login Page (Index.aspx), allowing them to securely log out of the application and requiring reauthentication for subsequent access.



#### **7.4.4.2 Home Page**

The Homepage, represented by HomeLogin.aspx, serves as a central hub for users to navigate between different modules and functionalities within the ERP software.

Buttons on the homepage are strategically placed to provide easy access to various sections of the ERP software, such as Employees, Stocks, Suppliers, Purchases, Receives, and Sales. Each button is linked to a specific page within the application, enabling seamless navigation. Bootstrap framework is implemented to style and structure the buttons on the homepage. Bootstrap's grid system and predefined button styles ensure consistent layout and appearance across different screen sizes and devices.



#### **7.4.5 List of names of all files**

- Index.aspx
- HomeLogin.aspx
- EmployeeForm.aspx
- EmployeeView.aspx
- Stocks.aspx
- StocksView.aspx
- SuppliersForm.aspx

- SuppliersView.aspx
- Sales.aspx
- SalesView.aspx
- PurchaseForm.aspx
- PurchaseView.aspx
- ReceiveForm.aspx
- ReceiveView.aspx

## **7.5 Backend Code/Design**

### **7.5.1 SQL Connection and Libraries**

The code utilizes the System.Data.SqlClient library to interact with the SQL Server database. Within each code file, a SqlConnection object is instantiated to establish a connection to the database. This connection string specifies the database server, database name, and authentication credentials required for accessing the database. The Open () and Close () methods are used in the methods assigned to the buttons to open and close the database connection, respectively.

### **7.5.2 Functions and Calling**

In the code, Object-Oriented Programming (OOP) principles are followed, where classes encapsulate data and behaviour. Each .cs file represents a class that encapsulates specific functionalities related to the ERP project, such as managing employees, purchases, receiving items, managing stocks, and handling suppliers. Private void functions serve as internal helpers, highlighting implementation details and internal logic specific to the class. They are not meant to be accessed directly from outside the class. Instead, these functions are utility methods that assist in performing specific tasks within the class, enhancing code readability and maintainability by abstracting complex operations.

On the other hand, public void functions are designed to be accessible from outside the class, making them suitable for invocation externally. Typically, these functions serve as event handlers or methods triggered by user interactions or system events. They represent actionable behaviors that can be invoked by external entities, such as user interfaces or other components of the system. By exposing these functions publicly, the class provides a clear interface for interacting with its functionalities, promoting modularity and extensibility in the codebase.

Private bool functions are integral components utilized across all pages within the code to perform specific checks or validations as they validate the existence of records through a count (\*) code implemented. These functions are designed to return a Boolean value, indicating the outcome of the validation process. For instance, the VendorcodeExist function is responsible for checking the existence of a vendor code in the database, ensuring uniqueness. Similarly, the

HasForeignKeyConstraints function verifies whether any foreign key constraints are associated with a particular vendor code, preventing its deletion if constraints exist.

Another example is the BarcodeExist function, which determines the presence of a barcode within the system, often employed for validation purposes during item processing workflows. Likewise, the InvoiceExist function serves to ascertain whether

an invoice number already exists in the system, maintaining uniqueness across transactions. Additionally, the EmployeeExist function validates the presence of an employee based on their unique identifier.

These private bool functions encapsulate specific validation or verification tasks essential for ensuring data integrity and enforcing business rules within the application. By incorporating these checks, the system can validate input data, prevent duplicate entries, and uphold referential integrity, thereby promoting the reliability and accuracy of operations performed across various functionalities. The functions mention above are mainly responsible and placed to prevent errors through if else conditions.

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private bool PORecord(string PONumber, String VendorCode, string RefNo)
{
    using (SqlConnection con = new SqlConnection(@"Data Source=LAPTOP-U47QFCE\MSSQLSERVER03;Database=ERP; User Id=sa1;Password=1234;"))
    {
        con.Open();
        SqlCommand cmd = new SqlCommand("SELECT COUNT(*) FROM PurchaseOrder WHERE PONumber = @PONumber AND VendorCode = @VendorCode AND RefNo = @RefNo", con);
        cmd.Parameters.AddWithValue("@PONumber", PONumber);
        cmd.Parameters.AddWithValue("@VendorCode", VendorCode);
        cmd.Parameters.AddWithValue("@RefNo", RefNo);
        int count = (int)cmd.ExecuteScalar();
        return count > 0;
    }
}
```

```
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protected void Get_Click(object sender, EventArgs e)
{
    Det.Open();
    try
    {
        if (string.IsNullOrEmpty(PONumber.Text) ||
            string.IsNullOrEmpty(VendorCode.Text) ||
            string.IsNullOrEmpty(RefNo.Text))
        {
            ErrorLabel.Text = "Please specify PO, Vendor and Ref!";
        }
        else if (!PORecord(PONumber.Text, VendorCode.Text, RefNo.Text))
        {
            ErrorLabel.Text = "Purchase not completed! Enter Correct PO, Vendor and Ref ";
        }
        else if (Receiveorder(PONumber.Text, VendorCode.Text, RefNo.Text))
        {
            SqlCommand cmd = new SqlCommand("Select BarCode, Description, UnitPrice, OrderQty, VendorName, Coeff, SUM(ReceivingQty) OVER (Partition
            cmd.Parameters.AddWithValue("@PONumber", PONumber.Text);
            cmd.Parameters.AddWithValue("@VendorCode", VendorCode.Text);
            cmd.Parameters.AddWithValue("@RefNo", RefNo.Text);

            using (SqlDataReader reader = cmd.ExecuteReader())
            {
                // ...
            }
        }
    }
}
```

### **7.5.3 JavaScript Code generation**

In the ERP project, JavaScript code generation from the frontend upon clicking typically involves dynamically updating or modifying the client-side behavior of the web application based on user interactions. This process often occurs in response to specific events triggered by the user, such as button clicks or form submissions, and is facilitated by the ASP.NET framework.

When a user interacts with the frontend interface by clicking buttons or performing actions, corresponding event handlers are triggered on the server-side backend. These event handlers, implemented in the code-behind files (e.g., .aspx.cs files), respond to the user's actions and execute specific server-side logic.

Within these event handlers, developers can generate JavaScript code dynamically by emitting JavaScript statements or functions as part of the server-side response. This can involve constructing JavaScript functions, modifying the attributes or behavior of HTML elements, or even executing JavaScript code directly.

#### **7.5.4 Data Retrieval for Update Purposes**

The "Edit" and "Update" buttons on each form is responsible in enabling users to interact with and modify existing data entries within the system. When a user clicks the after inserting the relevant data required "Edit" button, it triggers the retrieval of specific record details from the database. These details are then populated into the respective input fields, allowing users to view and modify the existing information conveniently.

Following the retrieval of data, users have the opportunity to make necessary changes or updates to the displayed information directly within the input fields. Once the modifications are complete, users can proceed to click the "Update" button. This action triggers the update operation, wherein the edited data is submitted back to the database.

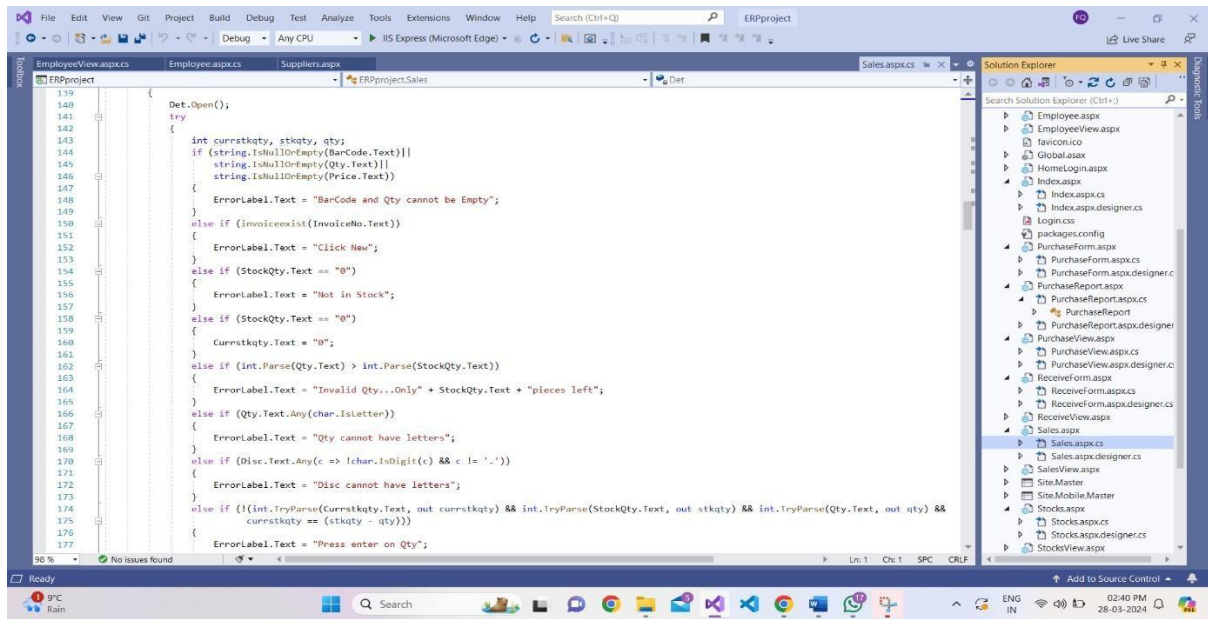
The "Update" button ensures that any modifications made by users are accurately saved and reflected in the database, thus maintaining data consistency and integrity across the system. Together, the "Edit" and "Update" buttons provide users with a streamlined and efficient means of editing and updating existing records, enhancing the overall usability and functionality of the system.

#### **7.5.5 Error Handling with if-else and try-catch**

Error handling in the code is primarily implemented using if-else statements for basic conditional checks and try-catch blocks for handling exceptions more comprehensively. Throughout the code, if-else statements are used to validate user inputs, check for specific conditions, and determine the flow of execution based on various criteria. For instance, if-else statements are utilized to ensure that required fields are filled, validate email formats, verify phone numbers, and check for the existence of certain data entries before proceeding with operations like insertion, deletion, or updates. These if-else statements help in preventing errors and ensuring that the code behaves as expected under different scenarios.

Additionally, try-catch blocks are employed to handle exceptions that may occur during the execution of code, such as database connection errors, SQL query failures, or unexpected runtime errors. By wrapping certain code segments within try blocks and catching exceptions in corresponding catch blocks, the application can gracefully handle errors without crashing or disrupting the user experience.





### **7.5.6 Generation of New and Unique IDs**

In the code files, the generation of new and unique IDs is achieved through various methods. For example, in the Purchaseform.aspx.cs file, the `ddlSupplier_SelectedIndexChanged` event handler generates a unique ID for each new purchase entry by concatenating the selected supplier's code with other details like product name, quantity, and price. Additionally, the `New_Click` event handler in the Sales.aspx.cs file increments the invoice number to generate a new and unique invoice ID for each sales transaction.

### **7.5.7 Insertion and Deletion operations**

In the code across all pages, insertion and deletion operations are implemented for managing data within the ERP system using the button present on each form “Save” and “Delete”. When it comes to insertion, new records are added to the respective database tables through events like button clicks. These operations are safeguarded by parameterized SQL queries, which inject values into the database, ensuring security and preventing SQL injection vulnerabilities. Additionally, validation checks are commonly implemented to verify the integrity and validity of the data being inserted.

On the other hand, deletion operations are responsible for removing specific records from the database tables, often triggered by user requests or system requirements. These operations are typically executed through event handlers like button clicks or administrative actions. Similar to insertion, deletion operations are protected using parameterized queries to prevent unauthorized access and maintain data integrity. Furthermore, before executing deletion operations, the code frequently performs validation checks to confirm the legitimacy of the deletion request and ensure compliance with any constraints or business rules.

As soon as the task is completed and no error is encountered a dialogue prompt on top right corner stating “Task Successfull”.

#### **7.5.8 Viewing Inserted Data Page**

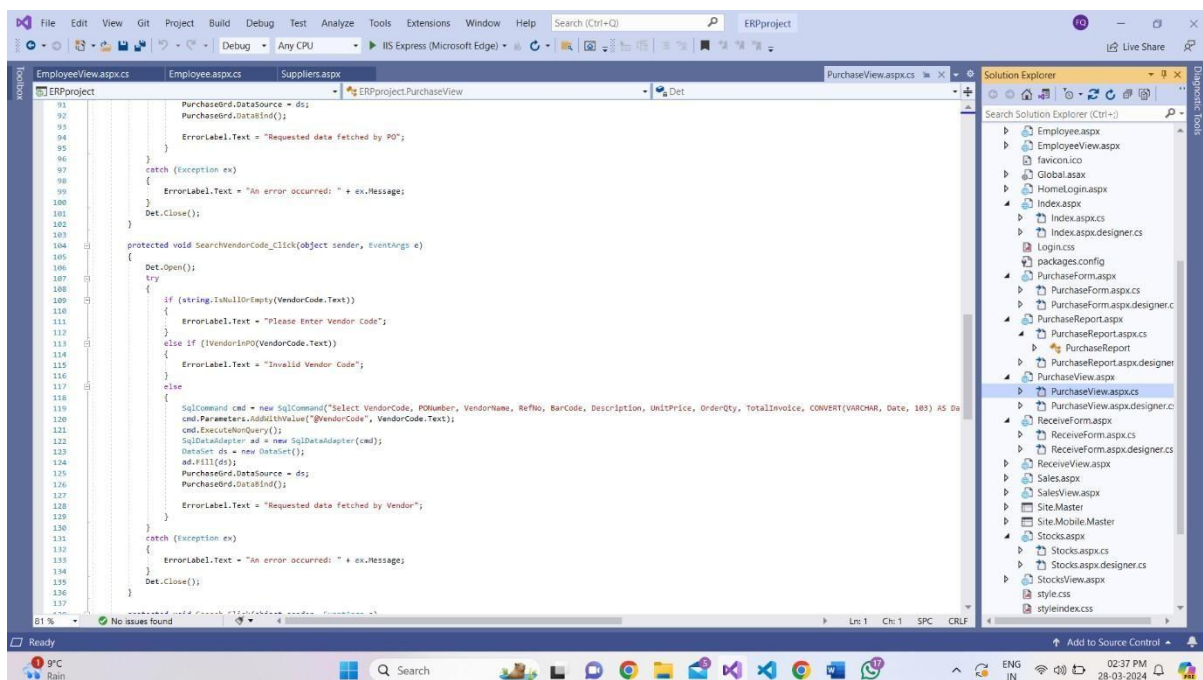
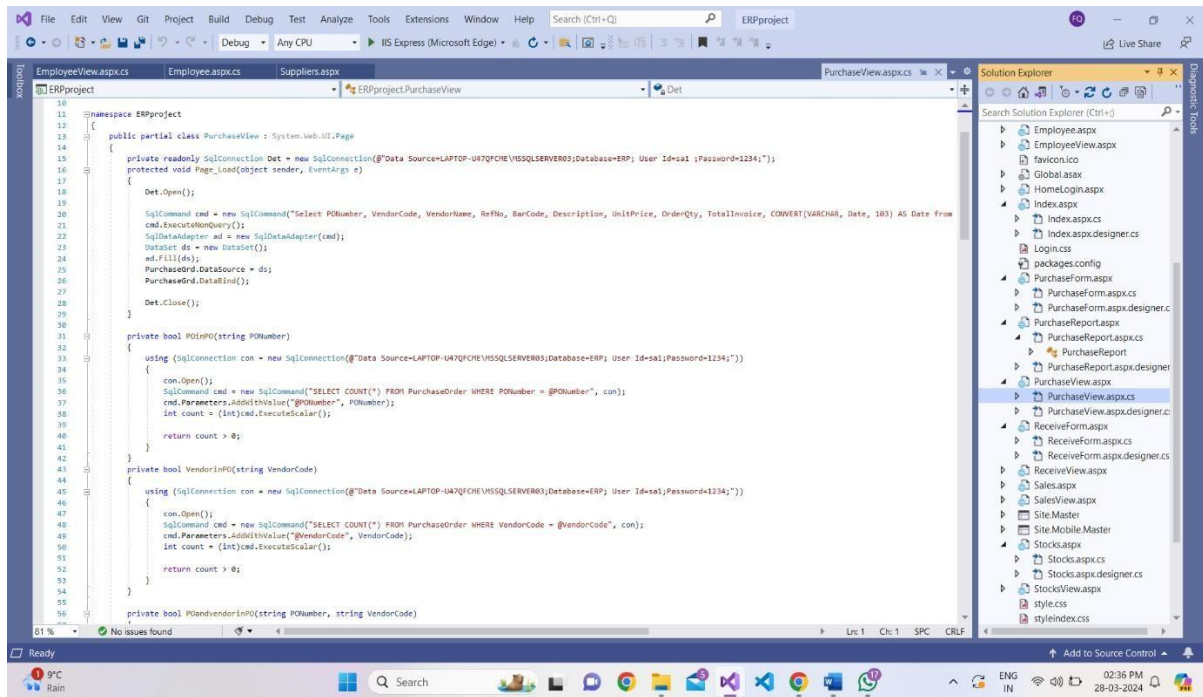
To view the data inserted through the functionalities described above, users can navigate to the corresponding pages in the ERP project's web application. For example, after inserting purchase details using the form, users can navigate to the Purchase view page to view the inserted records. The ASP.NET pages associated with each functionality display the relevant data retrieved from the database, providing users with a comprehensive view of the inserted data.

The code starts by establishing a connection to the SQL Server database using connection strings. This connection enables the application to retrieve and manipulate data stored in the database.

Data retrieval and presentation are handled through SQL queries. The code fetches data from the database tables based on user requests or predefined criteria. This retrieved data is then formatted and presented to users through data-bound controls like GridView or DataGrid, ensuring easy visualization and understanding. User interaction is managed through event handlers implemented to handle actions such as button clicks or search requests. These event handlers trigger specific actions like querying the database for employee details, sales records, received orders, purchase orders, stock information, or supplier data based on user input.

Robust error handling mechanisms are incorporated to catch and manage exceptions that may arise during database operations or data retrieval processes. This ensures that users receive informative error messages in case of any issues. A glimpse is provided below by the figure.





### 7.5.9 Login Page

The backend code for Login page is structured within the ERPProject namespace and encapsulated in the Index class, indicating its role as the code-behind file for the "Index.aspx" web page. Within the code, a SQL connection (SqlConnection) named con is established to interact with the SQL Server database. This connection string includes details such as the server's name, database name, and authentication credentials required for access.

In the Page\_Load method, which executes when the associated web page loads, no specific actions are defined. This method remains empty, suggesting that no initialization or processing tasks occur during the page load event.

The core functionality of the code lies in the Login\_Click method, triggered upon clicking the login button on the web page. Within this method, user input for the username and password is validated to ensure they are not empty. If either field is empty, an appropriate error message is displayed. Subsequently, a SQL query is executed to check the provided credentials against the database. If a matching record is found, the user is redirected to the "HomeLogin.aspx" page, indicating successful authentication. Otherwise, an error message notifies the user of incorrect credentials.

#### **7.5.10 Code implemented on Report formulating page**

This page serves a similar purpose to the retrieval of data in view pages, employing nearly identical code for data validation and retrieval. The key distinction lies in the absence of a GridView, replaced instead by a report generated using labels. Data is retrieved based on unique IDs provided in the text boxes at the top of the page.

#### **7.5.11 List of names of all files**

- Index.aspx.cs
- HomeLogin.aspx.cs
- EmployeeForm.aspx.cs
- EmployeeView.aspx.cs
- Stocks.aspx.cs
- StocksView.aspx.cs
- SuppliersForm.aspx.cs
- SuppliersView.aspx.cs
- Sales.aspx.cs
- SalesView.aspx.cs
- PurchaseForm.aspx.cs
- PurchaseView.aspx.cs
- PurchaseReport.aspx.cs
- ReceiveForm.aspx.cs
- ReceiveView.aspx.cs

## **8. Testing of the software**

### ***Why usability testing is important?***

Usability testing allows to learn things about user behaviour, needs, and expectations upfront. The insights and benefits gleaned from usability testing are essential to informing product development. Observing the way someone interacts with your product interface provides you with an opportunity to course correct your design and set development priorities from the get-go. UX blogger Sarah Elson, from the product growth team at LambdaTest, sums up the connection between usability testing and UX in her article about web development last year: "User experience is one of the most important if not the most

important aspect of making an online business successful. And that's what is measured under the scope of usability testing."

Whether you're building a website, an app or any other digital product, the principles behind usability testing hold true. *(Sarah elson, 2019)*

The general principles by Jakob Nielsen were utilized during the usability testing the principles were also known as heuristics listed below.

- **Visibility of System Status** • **User Control and Freedom** • **Error Prevention** • **Minimalist Design** • **Help Users Recognize, Diagnose, and Recover from Errors** • **Documentation**
- (Nielsen. J, 2019).

### **8.1 Visibility of the System Status**

The design should always keep users informed about what is going on, through appropriate feedback on the Page.

During the software testing phase, a comprehensive assessment was conducted, focusing primarily on the frontend of the website, which encompasses the elements visible to users. This evaluation aimed to ensure an intuitive and user-friendly interface, facilitating seamless interaction with the software. Key areas assessed included page navigation, visual design, consistency and branding, accessibility, user feedback mechanisms, responsive design, and performance. The effectiveness of page navigation was examined to ensure users could easily identify their current location and navigate to other pages. Visual design elements, such as font size, colour scheme, and layout, were analysed for readability, contrast, and overall aesthetic appeal. Consistency across pages and adherence to branding guidelines were crucial for maintaining a cohesive user experience. Accessibility features were reviewed to ensure inclusivity for users with disabilities, while feedback mechanisms kept users informed about their actions and the system's response. Responsive design ensured a consistent user experience across different devices, and performance optimization aimed to enhance loading times and interactions. By assessing these frontend aspects, the software aimed to optimize usability, engagement, and overall user satisfaction, with further usability testing sessions providing qualitative feedback on user experience.

### **8.2 User Control and Freedom**

Users often perform actions by mistake. They need a clearly marked "emergency exit" to leave the unwanted action without having to go through an extended process.

During the assessment, particular attention was paid to features that facilitate user control and freedom, including the implementation of undo functionalities, testing of navigation buttons for seamless task reversal, and the presence of clear and informative feedback labels. Users were encouraged to utilize familiar keyboard shortcuts such as Ctrl+Z for undoing actions, providing them with a quick and intuitive method to revert unintended changes. Navigation buttons and functionalities were rigorously tested to ensure users could effortlessly backtrack to their previous tasks or pages, promoting a sense of control and freedom within the software environment. Additionally, the

visibility and clarity of feedback labels were evaluated to ensure they effectively communicated the outcome of user interactions, enabling users to make informed decisions and take appropriate actions.

### **8.3 Error Prevention**

Good error messages are important, but the best designs carefully prevent problems from occurring in the first place. Either eliminate error-prone conditions, or check for them and present users with a confirmation option before they commit to the action.

During the assessment, pages were thoroughly tested by intentionally inserting inappropriate data in text boxes or leaving them blank and then clicking the corresponding buttons to trigger actions. This approach allowed for the evaluation of error prevention mechanisms within the software. Specifically, the assessment focused on whether the error label was triggered when users attempted to commit actions with invalid or missing data. Additionally, the content and clarity of the error messages displayed were assessed to ensure they effectively communicated the nature of the error and provided guidance on resolving it. By systematically testing these scenarios, the software's ability to prevent errors and guide users towards appropriate actions was evaluated,

### **8.4 Minimalist Design**

Interfaces should not contain information that is irrelevant or rarely needed. Every extra unit of information in an interface competes with the relevant units of information and diminishes their relative visibility.

During this stage of assessment for the ERP software, the assessor focused on identifying and evaluating the relevance of information presented in the user interface. This involved scrutinizing every element displayed on the interface to determine if it was essential for users to accomplish their tasks effectively. To assess this aspect, the assessor examined each component of the interface, including text, buttons, menus, forms, and other elements, to determine if they served a clear purpose aligned with user needs.

### **8.5 Help Users Recognize, Diagnose, and Recover from Errors**

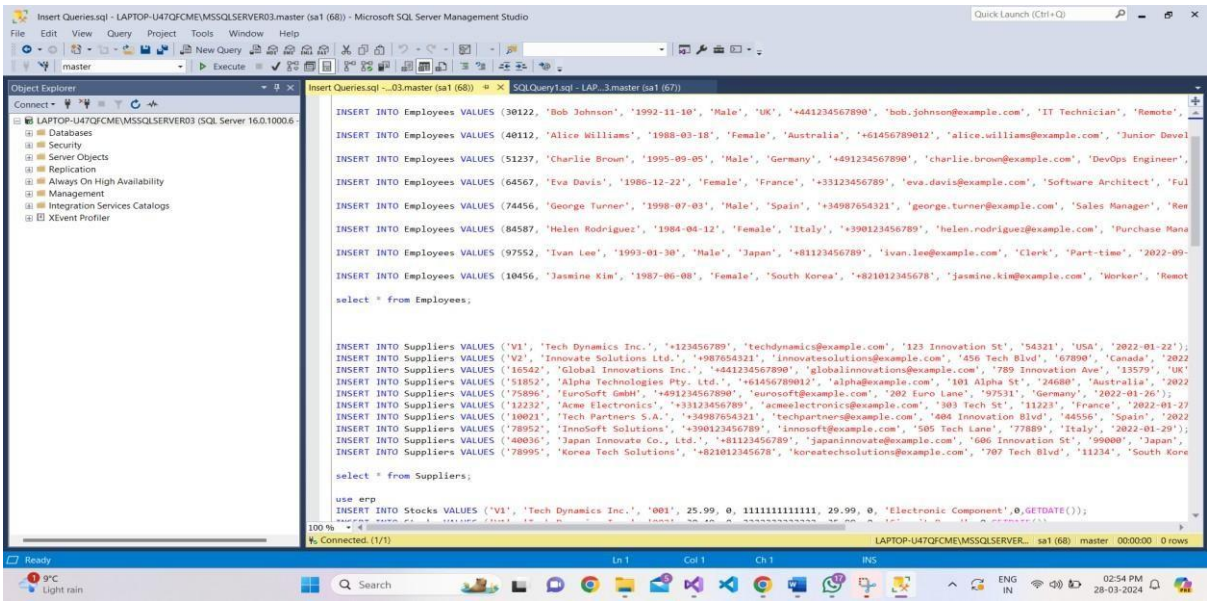
Error messages should be expressed in plain language (no error codes), precisely indicate the problem, and constructively suggest a solution.

The error messages were scrutinized to ensure they were expressed in clear and straightforward language, devoid of any technical jargon or cryptic error codes. This approach aimed to facilitate better understanding among users, minimizing confusion and frustration. The precision of error messages in identifying and articulating the encountered issue was thoroughly examined. Each message was evaluated to ensure it accurately pinpointed the problematic areas, such as specific input fields or actions, enabling users to comprehend the nature of the problem swiftly. The assessment encompassed the effectiveness of error messages in providing constructive solutions or guidance to users. Messages were analysed to ascertain whether they offered actionable

steps or suggestions for rectifying the error, empowering users to navigate through the issue seamlessly.

### 8.6 Data Integration

Dummy data was inserted directly into the database using SQL commands to meticulously assess the database's performance and its ability to handle data according to the predefined schema and constraints. This method was instrumental in validating the integrity of the database structure and identifying any potential errors or anomalies that may arise during data insertion and retrieval processes. Similarly, dummy data insertion through each frontend form was executed with precision and thoroughness, serving as a pivotal aspect of functionality testing. This seamless integration of data input via forms ensured that the ERP software effectively captured and processed user inputs, thereby validating the core functionality of the system.



### 9. Results for Manual Testing

Based on the website assessment conducted, several observations were made regarding different aspects of the interface and functionality. The evaluation highlighted concerns about the font size, which although is readable, appeared small, potentially impacting user experience by making it difficult for users to read text comfortably. This observation aligns with the assessment's focus on visual design elements such as font size, indicating the need for adjustments to enhance readability.

Furthermore, the assessment revealed issues with the functionality of two report forms, "Receive Report" and "Stock Report," which were found to be non-functional and absent from the "View Data" pages. This observation directly relates to the evaluation of page navigation and accessibility, indicating shortcomings in providing users with the ability to retrieve essential data effectively.

Another observation from the assessment was related to user interaction difficulties, particularly regarding data retrieval data using dates was not seen as an option and encountering irrelevant buttons such as "Contact" and "Performance" in the navigation.

These findings underscore the importance of assessing user feedback mechanisms and ensuring consistency across pages to enhance usability and streamline navigation.

Moreover, concerns were raised about the visual effects of the website, including a dark wallpaper and slightly transparent buttons, which although visible, detracted from the overall appeal. Issues such as textboxes defined in black color against a dark background further compounded visibility challenges, highlighting the importance of evaluating visual design elements for clarity and accessibility.

Error messages were also observed during the assessment, revealing issues with font size and positioning, which affected their visibility and effectiveness in communicating error notifications to users. This observation emphasizes the need to refine error message presentation to ensure clear communication of system status and user feedback.

Additionally, complexities in the purchase order page and challenges with the usability of the sales page were noted during the assessment. These observations highlight areas where user control and freedom could be enhanced by streamlining workflows and reducing the cognitive load on users.

Despite these challenges, the assessment acknowledged the effective operation of insertion, update, delete, and edit functions across forms, as well as the precision of calculations on "Receive Order," "Sales," and "Stocks," indicating positive aspects of the software's functionality in between the areas for improvement identified during the assessment.

## **10. Areas for further work**

### **10.1 Designing Part through CSS Optimization**

Enhancing the software's design through CSS optimization is imperative to improve its visual appeal and user experience. This involves refining the layout, color schemes, typography, and overall visuals of the interface. By optimizing CSS, we can ensure consistency in design elements across the entire software, making it visually appealing and intuitive for users. This process also includes enhancing responsiveness to different screen sizes and devices, thereby catering to a wider audience.

### **10.2 Non-Functional Pages like Receive Report and Stock Report**

Several pages, such as the receive report and stock report, are currently non-functional and require further development to make them operational. This entails identifying and resolving any underlying issues or bugs that prevent these pages from functioning as intended. Once addressed, thorough testing is necessary to ensure that these pages accurately retrieve and display the required data, providing users with valuable insights into received items and available stock.

### **10.3 Enhancing Data Retrieval Techniques in View Forms**

The view forms currently employ minimal data retrieval techniques, limiting their effectiveness in providing users with comprehensive data insights. Enhancements are needed to improve data retrieval techniques, such as optimizing database queries, implementing caching mechanisms, or utilizing more efficient data retrieval methods. By

enhancing data retrieval techniques, we can ensure that users have access to timely and accurate information, enhancing the software's usability and utility.

#### **10.4 Creating Triggers and Views in the Database**

The creation of triggers and views in the database is essential for enhancing data management capabilities and improving overall system performance. Triggers can be used to enforce data integrity constraints, automate repetitive tasks, or implement business logic at the database level. Views provide a virtual representation of data from one or more tables, allowing for simplified data access and manipulation. Creating triggers and views in the database can streamline data processing workflows and improve the efficiency of data retrieval operations.

#### **10.5 Granting Admin and Employee Privileges on the Login Page**

Admin privileges and employee privileges need to be granted on the login page to ensure secure access to different functionalities based on user roles. Admin users should have access to administrative features, such as user management, configuration settings, and system monitoring tools. Employee users, on the other hand, should have access to specific modules or functionalities relevant to their roles, such as data entry, reporting, or task management.

#### **10.6 Creating Contact Page and Implementing Performance Monitoring**

The creation of a contact page provides users with a means to communicate with the software's administrators or support team, enabling them to seek assistance, report issues, or provide feedback. Additionally, implementing a performance monitoring system for employees allows administrators to track employee productivity, assess performance metrics, and identify areas for improvement. By monitoring employee performance, administrators can ensure optimal utilization of resources and enhance overall organizational efficiency.

#### **10.7 Adding Warehouse Form and Tracking Transactions**

The warehouse form enables users to record and manage inventory-related information, such as item quantities, locations, and transactions. By tracking transactions within the warehouse, users can monitor stock movements, track inventory levels, and generate reports to optimize inventory management processes. As this is related to goods, Purchase and receive forms.

#### **10.8 Automated Testing**

Implementing automated testing procedures is essential to validate the software's functionality, performance, and reliability. Automated testing frameworks can help streamline the testing process by automating repetitive tasks, reducing manual effort, and ensuring consistent test coverage. By automating test cases, we can detect bugs and issues early in the development cycle, leading to faster feedback and more reliable software.

#### **10.9 Deployment of Software**

The deployment phase is critical for transitioning the software from development to production environments. It involves planning and executing the process of installing,

configuring, and making the software operational for end-users. Proper deployment strategies need to be devised to ensure a smooth and seamless transition, minimizing downtime and disruptions. This includes setting up servers, configuring databases, deploying necessary dependencies, and conducting thorough testing in the production environment before final release.

## **11. Conclusion**

In conclusion, this report has provided a comprehensive overview of the development process and evaluation of an ERP software solution. The introduction outlined the importance of ERP systems in modern business operations, setting the stage for the subsequent sections. The aims and objectives elucidated the specific goals and targets of the project, guiding the development and assessment phases.

The literature review delved into existing research and best practices in ERP system design and usability testing, informing the approach taken in this project. Building upon this foundation, the designing part provided information the user interface and backend architecture of the ERP software, emphasizing usability, functionality, and scalability.

The implementation section detailed the translation of design concepts into tangible software components, highlighting the technical intricacies involved in coding, integration, and system deployment. Following implementation, the testing phase rigorously evaluated the software's performance, usability, and adherence to design principles.

The results of manual testing provided valuable insights into areas of improvement, ranging from visual design enhancements to functional fixes. These findings culminated in a list of actionable items for further work, encompassing CSS optimization, functional page development, data retrieval techniques, database enhancements, user privilege management, and performance monitoring.

This project displays a systematic approach to ERP software development, encompassing design, implementation, testing, and iterative improvement. By addressing the identified areas for further work, the software aims to evolve into a robust, user-friendly solution that meets the diverse needs of modern enterprises.

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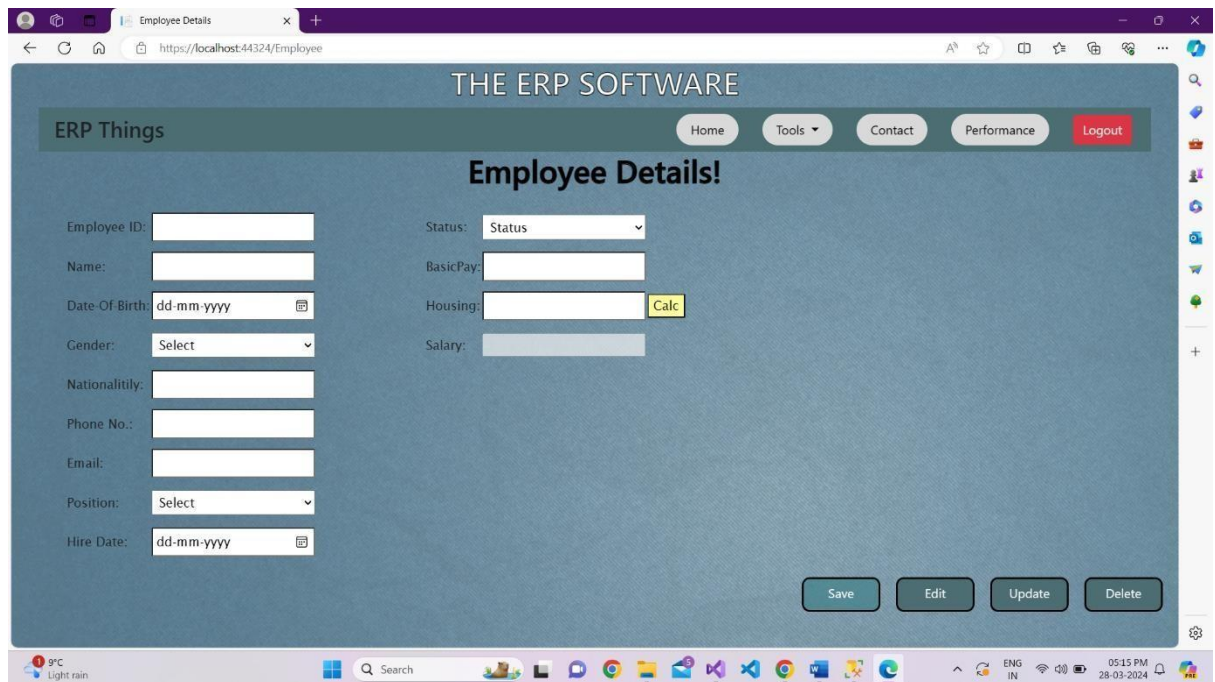
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## 13. Appendix



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### Employee Details!

Employee ID:  Status:

Name:  Basic Pay:

Date Of Birth:  Housing:

Gender:  Salary:

Nationality:

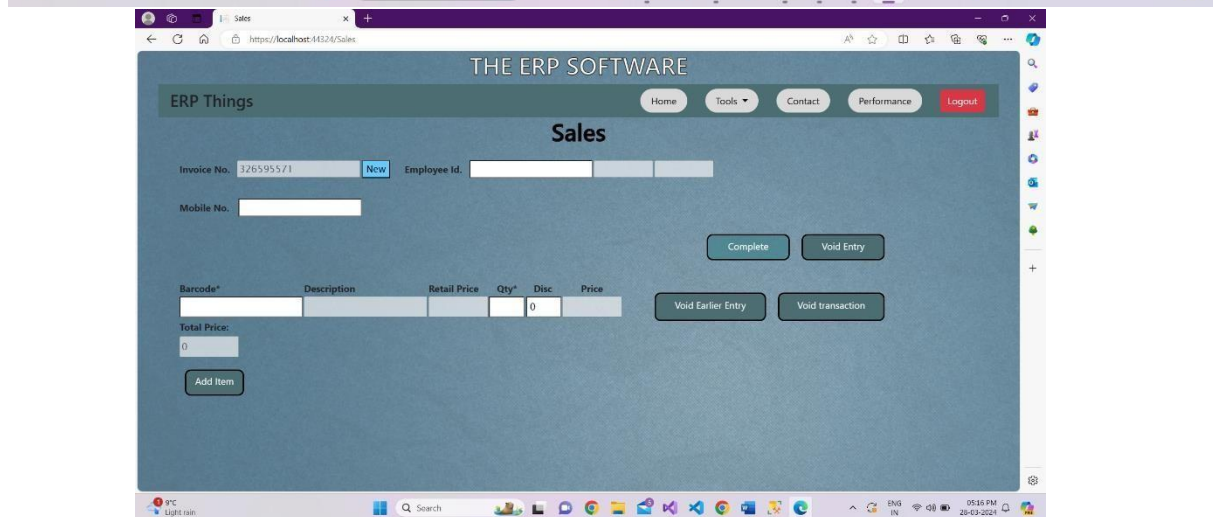
Phone No.:

Email:

Position:

Hire Date:

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### Sales

Invoice No.   Employee Id.

Mobile No.

| Barcode*             | Description          | Retail Price         | Qty*                           | Disc                 | Price                |
|----------------------|----------------------|----------------------|--------------------------------|----------------------|----------------------|
| <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text" value="0"/> | <input type="text"/> | <input type="text"/> |

Total Price:

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THE ERP SOFTWARE

ERP Things

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Receive Order

Employee Form »  
Stocks Form »  
Supplier Form »  
Purchase Goods »  
Receive Goods »  
Sales »

Received PO: 0  
Purchase View  
Purchase Report  
Calc

PO Number:

Invoice No.:

Receiving No.:

Vendor Code:

Ref No.:

Bar Code:

Vendor Name:

Description:

Order Qty:

Receive Qty:

Pending Qty:

Retail Price:

Total Invoice:

Receive Order

Delete Record

Clear

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Search

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THE ERP SOFTWARE

ERP Things

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Employee Details!

Invoice No.:

Search

| InvoiceNo | BarCode        | Description           | RetailPrice | Qty | Price  | Disc  | Date                |
|-----------|----------------|-----------------------|-------------|-----|--------|-------|---------------------|
| 32695566  | 555555555555   | Software Licenses     | 16.99       | 1   | 16.99  | 10.00 | 01-02-2024 02:35:31 |
| 32695566  | 11111111111111 | Electronic Component  | 29.99       | 1   | 26.93  | 10.20 | 01-02-2024 03:13:40 |
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| 32695566  | 11111111111111 | Electronic Component  | 29.99       | 1   | 26.93  | 10.20 | 01-02-2024 03:14:04 |
| 32695568  | 555555555555   | Software Licenses     | 16.99       | 1   | 15.29  | 10.00 | 01-02-2024 22:12:33 |
| 32695568  | 555555555555   | Software Licenses     | 16.99       | 3   | 45.87  | 10.00 | 01-02-2024 22:12:54 |
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| 32695572  | 555555555555   | Software Licenses     | 16.99       | 1   | 15.29  | 10.00 | 09-02-2024 01:03:14 |

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ERP Things

HomeToolsContactPerformanceLogout

Stocks!

Vendor Code:

Bar Code:

Description:

Vendor Name:

Retail Price:

Disc:

Ref No.:

Sold Qty:

Unit Price:

Stock Qty:

Save

Add

Stock

Add Disc

Clear

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Search

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